

Highlights

UniIA-DET: Illumination-Adaptive Detection for UAV-Based Wind Turbine Inspection

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- CL-WTBD is constructed as an illumination-balanced benchmark for UAV-based wind turbine blade defect detection under normal-light, backlight, and low-light conditions.
- UniIA-DET introduces a detector-supervised input-space correction module to improve defect detection robustness under complex illumination.
- IAPM can be integrated with CNN-, Transformer-, and SSM-based detectors; the primary YOLOv8n instantiation shows consistent gains under backlight and low-light conditions across three training seeds.

UniIA-DET: Illumination-Adaptive Detection for UAV-Based Wind Turbine Inspection

Xunyang Su^a, Chengjia Wang^b, Xinghui Dong^{a,*}

^a*State Key Laboratory of Physical Oceanography and the Faculty of Information Science and Engineering, Ocean University of China, Qingdao, 266100, China*

^b*School of Mathematical and Computer Sciences, Heriot-Watt University, Riccarton Mains Road, EH14 4AS, Edinburgh, United Kingdom*

Abstract

Automated defect detection for wind turbine blades remains challenging under uncontrolled outdoor illumination, where backlight and low-light conditions suppress texture cues, reduce local contrast, and increase false detections. We address this problem from both data and model perspectives. First, we construct a Complex-Lighting Wind Turbine Blade Defect Dataset (CL-WTBD) by integrating heterogeneous UAV-based inspection sources and harmonizing their annotations into a unified taxonomy. CL-WTBD is balanced across normal-light, backlight, and low-light domains, providing a controlled benchmark for evaluating illumination robustness in blade defect detection. Second, we propose UniIA-DET, an illumination-adaptive detection framework built on top of an off-the-shelf object detector. Instead of relying on a standalone image enhancement module or aligning features only in latent space, UniIA-DET introduces an Illumination-Adaptive Prompting Module (IAPM) before the detector backbone. The IAPM uses content-driven dynamic routing to compose an input-specific correction from a learnable prompt pool, which is additively injected into the raw image to calibrate exposure and contrast before feature extraction. The module is optimized jointly with the detector using the detection objective, allowing the correction process to be guided by classification and localization requirements

*Corresponding author

Email addresses: `suxunyang@stu.ouc.edu.cn` (Xunyang Su), `chengjia.wang@hw.ac.uk` (Chengjia Wang), `xinghui.dong@ouc.edu.cn` (Xinghui Dong)

rather than by perceptual reconstruction criteria. Experiments on CL-WTBD show that the proposed module improves robustness under harsh illumination. On the YOLOv8n instantiation, the main benefit of IAPM appears under harsh illumination: averaged over three seeds, it improves the Backlight mAP@50 from 0.352 to 0.392 and the Low-Light mAP@50 from 0.392 to 0.446, raising the macro-domain mAP@50 from 0.424 to 0.466, while the global mAP@50 changes from 0.387 to 0.398, within the run-to-run variation. Additional experiments with RT-DETR and Mamba-YOLO indicate that IAPM can be attached to different detector families without modifying their original architectures. These results demonstrate the practical value of detector-oriented input-space illumination correction for UAV-based wind turbine blade inspection.¹

Keywords: Wind turbine inspection; defect detection; illumination-adaptive; input-space correction; dynamic routing; complex-illumination benchmark

1. Introduction

Wind energy has expanded rapidly in installed capacity over the past decade as a key pillar of the global transition toward carbon neutrality [1, 2]. Wind turbine blades harvest wind power and directly determine energy conversion efficiency. In long-term outdoor operation, the blades are exposed to rain, sand, salt spray, and lightning, making them susceptible to structural degradations, such as surface cracks, erosion, and lightning damage [3]. If not detected early, small defects may grow quickly and eventually trigger severe failures, leading to costly downtime and maintenance [4]. To scale inspections, unmanned aerial vehicles (UAV) equipped with high-resolution cameras have become a practical standard for data acquisition, shifting the bottleneck from sensing to the automated interpretation of large volumes of aerial imagery with deep learning [5, 6].

¹Code, configurations, trained weights, dataset metadata, and split files will be released publicly upon acceptance. Image redistribution will follow the source-dataset licenses; details are provided in the Data availability statement.

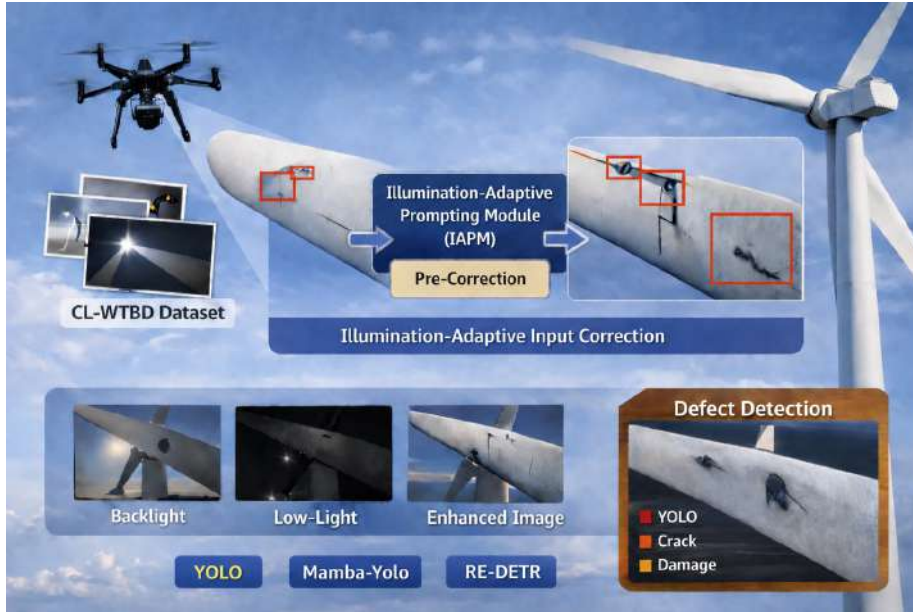


Figure 1: UAV-based blade inspection under harsh illuminations. Backlight and low-light conditions in CL-WTBD suppress texture cues and increase false negatives. We introduce an Illumination-Adaptive Prompting Module (IAPM) as a learnable calibration stage to mitigate illumination distortion before detection.

Although modern object detectors, such as the YOLO series [7] and Faster R-CNN [8], achieve excellent performance in controlled imagery, their reliability often degrades in field conditions where illumination is uncontrolled [9]. UAV-based inspections are frequently conducted across variable weather and sun angles, producing extreme visual interference. For example, strong backlight induces silhouette-like appearances that obscure surface textures, while low-light imagery suffers from severe noise and weak contrast. A common practice is to prepend an Image Enhancement (IE) module as a separate pre-processing step (e.g., histogram equalization or Retinex-based methods) [10, 11]. Such “enhance-then-detect” pipelines, however, are often mismatched with detection objectives, as discussed in Section 2.2. Progress is further constrained by data bias. That is to say, existing blade-defect datasets are dominated by normal-light images and provide limited coverage of backlight and low-light cases, weakening

the ability of a detector to generalize across illumination regimes [12].

To bridge these gaps, we address harsh-illumination inspection from both data and methodology perspectives. First, we collect a Complex-Lighting Wind Turbine Blade Defect Dataset (CL-WTBD). By integrating samples from heterogeneous sources, harmonizing their labels, and applying a minimum-alignment strategy for illumination-domain balancing, CL-WTBD enforces equal balance across normal, backlight, and low-light domains. The practical challenge and our design target are illustrated in Fig. 1.

Building upon this benchmark, we propose UniIA-DET, an illumination-adaptive detection framework built on top of an off-the-shelf object detector for UAV-based wind turbine inspection. In contrast to feature-space domain adaptation methods that mainly align deep representations, UniIA-DET follows a detector-oriented pre-correction principle by calibrating the input image before feature extraction. Specifically, it inserts an Illumination-Adaptive Prompting Module (IAPM) into the pixel input space. Conditioned on the global illumination context, the IAPM dynamically composes an input-specific correction from a learnable prompt pool and injects it additively into the raw image. This learnable correction alleviates exposure and contrast distortions before the first feature extraction stage, thereby enhancing weak defect cues in shadowed regions and suppressing interference caused by overexposure. Since the detector architecture itself is not modified, the proposed module can be attached to standard detection backbones in a lightweight manner. The conceptual distinction of UniIA-DET lies in reformulating illumination handling as a detector-supervised input calibration problem. Unlike generic image enhancement, which mainly optimizes visual quality, the proposed correction is learned through detection losses and is therefore shaped by classification and localization requirements. Unlike feature-level domain adaptation, UniIA-DET acts before feature extraction and preserves the original detector topology, allowing illumination correction to influence early defect-relevant feature formation without introducing an additional feature-alignment objective.

Rather than claiming a new correction paradigm, this work investigates

how detector-supervised illumination-conditioned calibration can improve defect detection robustness under controlled illumination variations. The main contributions can be summarized as follows.

- We construct CL-WTBD, a multi-source and illumination-balanced benchmark for UAV-based wind turbine blade defect detection. The benchmark provides an explicit illumination-domain assignment protocol and enables controlled evaluation under Normal-Light, Backlight, and Low-Light conditions.
- We propose UniIA-DET, a detector-supervised illumination-conditioned input-space calibration framework. The proposed IAPM dynamically composes correction patterns from a learnable prompt pool according to image-level illumination context and injects them before feature extraction.
- We conduct extensive experiments on representative detector families, adopting YOLOv8n [7] as the primary instantiation and evaluating architectural compatibility on RT-DETR [13] and Mamba-YOLO [14]. Averaged over three seeds, IAPM improves the mAP@50 by about +0.04 and +0.05 under backlight and low-light conditions and raises the macro-domain mAP@50 from 0.424 to 0.466, while the global mAP@50 stays within the run-to-run variation. Together with Grad-CAM observations, these results indicate consistent robustness gains under harsh lighting with limited additional overhead.

The remainder of this paper is organized as follows. Section 2 reviews related work on UAV-based blade inspection, object detection under adverse conditions, and visual prompting. Section 3 introduces CL-WTBD and details the multi-source integration and domain balancing strategy. Section 4 presents UniIA-DET, including the formulation of IAPM and the optimization objective. Section 5 reports experimental results. Section 6 concludes the paper and discusses future directions.

2. Related Work

2.1. Wind Turbine Blade Inspection

Wind turbine blade inspection has progressively shifted from manual assessment to UAV-assisted data collection and automated analysis [15, 16, 17]. Early pipelines mainly relied on handcrafted image features, combining operators such as Canny edges, wavelet descriptors, or Histogram of Oriented Gradients (HOG) with shallow classifiers to localize surface defects [18]. While effective in constrained settings, these approaches generalize poorly to UAV imagery, where viewpoint changes, complex backgrounds, and illumination fluctuations violate the assumptions behind fixed feature designs [19].

Deep learning has since become the dominant paradigm. Convolutional Neural Network (CNN) based detectors can be broadly grouped into two-stage and one-stage models. Two-stage models, such as Fast R-CNN [20] and Faster R-CNN [8], first generate region proposals and then perform classification and bounding-box regression for refinement, achieving strong localization accuracy but typically incurring higher inference latency [20]. One-stage detectors, such as SSD [21] and the YOLO family [7], formulate detection as dense prediction and are widely adopted for real-time industrial inspection due to their favorable accuracy–efficiency trade-off [21, 22]. In this work, we use YOLOv8 [7] as the primary detector instantiation, given its mature training ecosystem and stable reproducibility in practical UAV pipelines.

While YOLOv8 serves as our primary baseline, we also consider alternative architectures to evaluate architectural compatibility. Transformer-based detectors, such as DETR [23] and its efficiency-oriented variants, leverage attention mechanisms to capture long-range dependencies [13]. More recently, State Space Models (SSMs) [24], such as Mamba-style backbones [14], have been increasingly explored for sequence modeling with linear complexity, offering an alternative route to long-context representation learning [25]. Despite these advances, robustness in the field remains constrained by data bias. In particular, most blade-defect datasets are dominated by normal-light imagery, with backlight and

low-light cases severely underrepresented, resulting in long-tailed distributions and performance drops under real operating conditions [26, 27].

2.2. Object Detection in Harsh Environments

Illumination variations, including low-light conditions, backlight glare, haze, and fog, constitute one of the central challenges for outdoor object detection. Existing solutions commonly fall into two categories: image enhancement and domain adaptation.

Many pipelines apply image enhancement as an external pre-processing stage. Classical methods, including Histogram Equalization [28], CLAHE [29], and Retinex-based formulations [30], improve contrast through global or multi-scale normalization. Learning-based enhancers, such as LLNet [31] and Zero-DCE [32], further attempt to recover illumination using data-driven mappings or curve estimation [31, 32]. However, "enhance-then-detect" strategies are not always aligned with the detection objective. When an enhancer is optimized purely for perceptual quality, it may in some cases amplify sensor noise, distort local textures, or introduce artifacts that reduce the semantic separability required by detectors [33, 34, 35].

Domain Adaptation (DA) methods aim to reduce the distribution gap between a source domain, typically normal-light images, and a target domain, such as harsh-illumination images. Image-level translation approaches based on Generative Adversarial Networks (GANs) are frequently used for day-to-night style transfer and data augmentation [36]. Feature-level adaptation further aligns deep representations using adversarial objectives or gradient reversal mechanisms to encourage domain-invariant features [37]. However, these approaches can be computationally heavy and may compromise structural fidelity. For instance, image translation may hallucinate patterns or alter defect morphology, while feature alignment may suppress domain-specific cues that are informative for defect localization.

In contrast, our approach integrates illumination correction into the detection pipeline through a lightweight, trainable input-space correction mechanism.

From a detection perspective, the objective of illumination correction is different from that of generic image enhancement. Built-in camera auto-exposure and post-processing enhancement methods mainly aim to improve global visibility or perceptual image quality. In general, improving image quality is expected to benefit object detection, because clearer images can reveal weak structures, improve local contrast, and reduce ambiguous visual evidence. Our intention is not to argue against image enhancement. Rather, the issue considered here is more specific: when an enhancement module is optimized mainly for perceptual quality, visibility, or reconstruction fidelity, the resulting visual improvement may not always translate monotonically into better classification and localization performance for subtle industrial defects, as measured by detection AP. In particular, when the correction objective is not fully aligned with the detection objective, operations such as contrast or exposure adjustment may modify local textures, amplify noise, or shift the color and contrast patterns that are informative for ranking and localizing small defects. Therefore, for UAV-based blade defect detection, we focus on detection-oriented input calibration, where the correction is optimized directly by the classification and localization losses rather than by perceptual reconstruction criteria alone.

Motivated by this observation, we optimize the correction module jointly with the detector. In this way, the correction module receives supervision from classification and bounding-box regression losses, encouraging the learned calibration to preserve defect-relevant structures instead of merely improving visual appearance. This distinction is important for UAV-based blade inspection, where defects are often small, low-contrast, and sensitive to illumination-induced appearance changes. Therefore, the key question is not only how to make a harsh-illumination image visually clearer, but how to learn a correction that is directly beneficial for defect classification and localization. This motivates the detector-oriented input-space correction adopted in UniIA-DET.

IAPM is also related to near-input normalization, such as instance normalization, adaptive normalization, and input-level standardization, which can likewise be trained under detection supervision. These methods re-scale and

re-center the input using per-channel statistics, and the transform is spatially uniform within each channel. IAPM differs in two respects. First, it composes an illumination-conditioned correction from learnable prompt prototypes and injects the correction into the input space through a residual additive scheme. The generated correction is input-adaptive through dynamic routing, allowing the detector to receive illumination-aware calibration signals before feature extraction. Second, instead of rescaling each input by its own statistics, IAPM learns a shared pool of correction bases and combines them through content-driven routing. A quantitative comparison with three near-input normalization baselines is reported in Section 5.4.

2.3. Input-Space Prompting and Detector-Oriented Correction

Prompt learning was first popularized in Natural Language Processing (NLP) as an efficient way to adapt large pre-trained models to new tasks by optimizing a small set of prompt parameters rather than full fine-tuning [38]. This idea has been extended to vision, where visual prompting introduces learnable prompts at the input level or within network layers to steer a pre-trained backbone toward a target distribution or task [39].

In this work, we adopt the term “prompt” in an input-space calibration sense. The prompt in IAPM denotes a learnable correction basis that is dynamically composed for each input image according to its illumination context. Different from prompt tokens used in vision-language models or Transformer-based recognition models, IAPM does not modify the detector architecture or insert additional tokens into intermediate layers. It provides a lightweight calibration interface before feature extraction, so that illumination correction can be optimized jointly with the detection objective. Therefore, UniIA-DET is positioned as a detector-oriented input calibration method that uses prompt-based correction composition for illumination robustness.

A representative line of research is Visual Prompt Tuning (VPT), which inserts learnable prompt tokens into Vision Transformers (ViTs) to adapt to downstream datasets with a small number of trainable parameters [40]. Prompt-

ing mechanisms have also been explored in low-level vision tasks, such as image restoration and style manipulation, where prompts steer models toward desired appearance transformations [41, 42]. Self-supervised and generative anomaly detection methods have also been explored for visual and video anomaly modeling under limited abnormal samples [43, 44]. Most existing approaches rely on *static* prompts shared across all inputs, or target recognition settings where prompting is performed in the token/feature space. In contrast, *dynamic, pixel-level prompting* tailored to illumination distortions in industrial detection has received limited attention.

Instead of applying static prompts or relying on feature-space alignment [45], UniIA-DET addresses this gap by introducing an Illumination-Adaptive Prompting Module (IAPM) that generates input-conditioned correction prompts and injects them into the raw image through a residual additive scheme. This design treats prompting as detector-oriented input calibration rather than token insertion or generic visual enhancement. By optimizing the correction jointly with the detector, IAPM provides a lightweight pre-correction interface that can be evaluated with heterogeneous detector families, including CNN-, Transformer-, and SSM-based architectures.

3. The CL-WTBD Dataset

We collect CL-WTBD (Complex-Lighting Wind Turbine Blade Defect Dataset) as a benchmark for wind turbine blade defect detection under realistic UAV-based inspection conditions, where illuminations are largely uncontrolled. This section describes (i) multi-source integration and label harmonization, (ii) illumination domain assignment and balancing, and (iii) annotation statistics and dataset splits.

3.1. Data Integration and Label Harmonization

CL-WTBD is curated by consolidating UAV-based inspection images from four heterogeneous datasets, including DOC [46], ADO [47], DTU-Drone [48],

Table 1: Taxonomy standardization. Heterogeneous labels from DOC [46], ADO [47], DTU-Drone [48], and WTBDI [49] are consolidated into three macro-classes to ensure semantic consistency.

Macro-Class	Original Labels from Source Datasets
Crack	<i>Structural discontinuities:</i> Surface Crack, Crack
Damage	<i>Erosion & Degradation:</i> Erosion, Le-erosion, Pinholes <i>Coating loss & Peeling:</i> Peeling, Peeled paint, Flaking, Paint, Chip <i>Structural damage:</i> Damaged, Damaged or Misaligned, Deformity, Drain hole impairment <i>Impact & Burns:</i> Burning, Lightning strike
Dirt	<i>Contaminants & Attachments:</i> Dirt, Oil, Oil leakage, PU-tape <i>Appearance degradation:</i> Rusty, Discoloration

and WTBDI [49]. Directly merging these sources is non-trivial because their annotation vocabularies differ in naming conventions (e.g., “*Le-Erosion*” vs. “*Erosion*”) and semantic granularity (e.g., “*Peeling*” vs. “*Peeled Paint*”). Such inconsistencies would fragment categories and undermine statistical reliability.

To obtain a coherent benchmark, we harmonize all source labels into three mutually exclusive macro-classes, including Crack (structural discontinuities), Damage (physical surface damage that may impair aerodynamic performance, including material loss, coating degradation, deformation, or burn/strike traces),

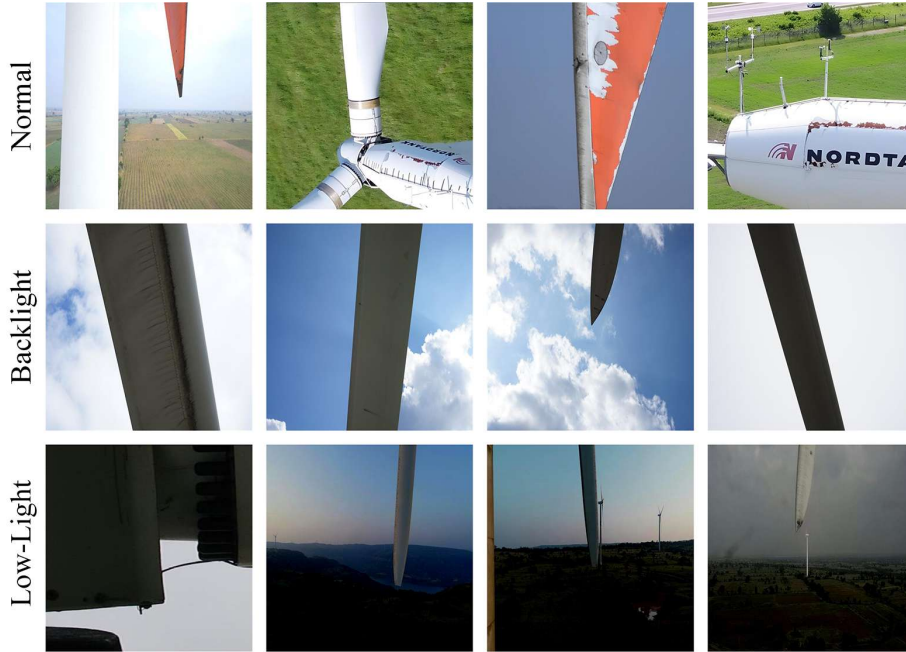


Figure 2: Representative CL-WTBD samples under three illumination domains. The three rows correspond to **Normal-Light** (top), **Backlight** (middle), and **Low-Light** (bottom), respectively.

and Dirt (non-structural attachments or appearance degradation). The mapping was reviewed by domain experts to ensure semantic correctness and to prevent conflating benign appearance changes with true defects. The resulting taxonomy standardization is summarized in Table 1.

It should be noted that the current detector focuses on coarse-level defect detection under complex illumination, rather than fine-grained diagnosis of specific defect subtypes. The three macro-classes are introduced to harmonize heterogeneous annotations from different source datasets and to support stable evaluation under complex illumination. However, this simplification inevitably reduces defect-specific information. For example, the Damage class contains multiple practical defect types, such as erosion, peeling, deformation, burning, and lightning-strike traces. As a result, the current setting does not support fine-grained recognition of individual defect subtypes. Fine-grained defect classi-

fication and severity assessment are regarded as important future extensions.

3.2. Illumination-Domain Assignment Protocol

To reduce the subjectivity of illumination-domain assignment, we first define a deterministic rule based on quantitative brightness statistics and then evaluate its agreement with independent human judgment.

Each RGB image is converted into the HSV color space, and the Value channel is used as the brightness representation. Given the Value channel $V \in \mathbb{R}^{H \times W}$, we compute the global mean brightness:

$$V_{\text{global}} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W V(i, j). \quad (1)$$

Since global brightness alone may not distinguish backlight images containing bright backgrounds and underexposed blades, we further estimate a coarse foreground-background partition using Otsu thresholding on the Value channel. The bright region above the Otsu threshold is treated as background (typically sky), while the darker region is regarded as foreground (typically the blade). Based on this partition, we compute the foreground brightness V_{fg} , background brightness V_{bg} , and foreground-background contrast:

$$\Delta V = V_{\text{bg}} - V_{\text{fg}}. \quad (2)$$

The illumination domain is assigned according to a fixed deterministic rule. Specifically, an image is categorized as Backlight when

$$V_{\text{bg}} > 160, \quad \Delta V > 70. \quad (3)$$

Otherwise, an image is categorized as Low-Light when

$$V_{\text{global}} < 85. \quad (4)$$

The remaining images are assigned to the Normal-Light domain.

The thresholds $T_{\text{bright}} = 160$, $T_{\text{contrast}} = 70$, and $T_{\text{low}} = 85$ were fixed as operating settings before illumination balancing. They were not selected

Table 2: Agreement between human and rule-assigned illumination labels in the 300-image verification sample. Rows denote human labels and columns denote rule-assigned labels. The sample contains 100 images from each rule-assigned domain.

Human label	Rule-assigned label			Total
	Normal-Light	Backlight	Low-Light	
Normal-Light	90	8	5	103
Backlight	7	91	3	101
Low-Light	3	1	92	96
Total	100	100	100	300

through a recorded optimization procedure. Before balancing, the candidate pool contained 2,955 images, including 1,412 Normal-Light images, 812 Backlight images, and 731 Low-Light images. No threshold adjustment was performed after balancing. The final CL-WTBD benchmark was constructed using minimum-alignment balancing, resulting in 731 images per illumination domain and 2,193 images in total. Threshold sensitivity was subsequently examined by varying each threshold independently by ± 10 while keeping the other two fixed.

To assess agreement between the deterministic assignment rule and human judgment, we conducted an independent blind annotation study on a stratified random subset of 300 images from the complete candidate pool, with 100 images sampled from each rule-assigned illumination domain. The evaluator was not involved in dataset construction and had no access to the rule-assigned labels or threshold values. Table 2 presents the confusion matrix, with human labels as rows and rule-assigned labels as columns. The labels agreed for 273 images, giving 91.0% agreement in this rule-stratified verification sample. With equal rule-assigned class marginals, the expected agreement was $1/3$ and Cohen’s κ was 0.865. Of the 27 disagreements, 15 occurred between Normal-Light and Backlight, eight between Normal-Light and Low-Light, and four between Backlight and Low-Light.

Table 3: Sensitivity of the illumination-domain partition and the resulting balanced dataset size to threshold perturbations on the complete 2,955-image candidate pool before balancing. Each threshold was varied independently by ± 10 , while the other two thresholds were kept at their default values ($T_{\text{bright}} = 160$, $T_{\text{contrast}} = 70$, and $T_{\text{low}} = 85$). The reported domain counts are the rule-assigned counts before balancing, given in the order Normal-Light/Backlight/Low-Light. Reassigned images denote images whose rule-assigned domain differs from the default partition. For each setting, the balanced dataset size was computed as $N_{\text{balanced}} = 3 \min(N_{\text{NL}}, N_{\text{BL}}, N_{\text{LL}})$. The change rate was calculated relative to the default balanced size of 2,193 images.

Setting	Rule-assigned counts (NL/BL/LL)	Reassigned images (count, %)	Balanced N	Change rate
Default	1,412/812/731	–	2,193	+0.00%
$T_{\text{bright}} = 150$	1,335/889/731	77 (2.6%)	2,193	+0.00%
$T_{\text{bright}} = 170$	1,482/742/731	70 (2.4%)	2,193	+0.00%
$T_{\text{contrast}} = 60$	1,347/877/731	65 (2.2%)	2,193	+0.00%
$T_{\text{contrast}} = 80$	1,478/746/731	66 (2.2%)	2,193	+0.00%
$T_{\text{low}} = 75$	1,463/812/680	51 (1.7%)	2,040	-6.98%
$T_{\text{low}} = 95$	1,331/812/812	81 (2.7%)	2,436	+11.08%

To evaluate the stability of the illumination-domain partition, we independently perturbed each threshold by ± 10 on the complete candidate pool while keeping the other two thresholds fixed at their default values. Table 3 reports the domain counts, label reassignments relative to the default partition, and the calculated dataset size after minimum-alignment balancing.

Across the six threshold perturbations, 51–81 images (1.7%–2.7% of the candidate pool) change illumination labels. After minimum-alignment balancing, the calculated benchmark size ranges from 2,040 to 2,436 images, compared with 2,193 under the default thresholds, corresponding to changes of -6.98% and $+11.08\%$, respectively. These changes arise from varying T_{low} , which changes the smallest-domain count. The tested perturbations of T_{bright} and T_{contrast} leave the balanced size unchanged, although the retained image set may differ. Thus, limited label reassignment before balancing does not imply comparable stability in the size or composition of the balanced benchmark.

3.3. Illumination-Domain Balancing Strategy

Operational UAV-based inspection data typically follow an imbalanced illumination distribution, with normal-light images dominating and harsh-illumination cases being relatively rare. After assigning illumination-domain labels using the protocol described above, we apply a minimum-alignment strategy to support a controlled evaluation of illumination robustness. Specifically, larger illumination domains are down-sampled to match the smallest domain. For the default construction, the target size was 731 images per illumination domain. Image identifiers were sorted within each illumination domain. Using random seed 42, we sampled 731 images uniformly without replacement from each of the Normal-Light and Backlight domains and retained all 731 Low-Light images. The resulting dataset contained 731 Normal-Light, 731 Backlight, and 731 Low-Light images, for a total of 2,193 images.

As a result, CL-WTBD is strictly balanced at a 1:1:1 ratio with **731 images per domain** (2,193 images in total), which are described as follows:

- **Normal-Light:** images with moderate global brightness and no extreme foreground-background brightness gap, where blade textures and defect cues remain clearly visible.
- **Backlight:** images with high background brightness and a large brightness contrast between the sky/background and the blade foreground, often leading to silhouette-like blade appearances and suppressed surface textures.
- **Low-Light:** images with low global brightness and generally underexposed foreground and background regions, usually accompanied by reduced contrast, low signal-to-noise ratio (SNR), or motion blur.

Representative samples from each domain are shown in Fig. 2.

3.4. Annotation Statistics and Splitting

CL-WTBD contains 2,193 images with 4,186 bounding-box annotated instances. Table 4 reports the category distribution under the train/validation/test

Table 4: CL-WTBD statistics. Illumination domains are equally balanced, while defect categories preserve the naturally imbalanced distribution.

Category	Train Inst.	Val Inst.	Test Inst.	Total Inst.
Crack	358	45	44	447
Damage	2,284	286	286	2,856
Dirt	707	88	88	883
Total	3,349	419	418	4,186

split. While the illumination domains are balanced by construction, the defect categories retain the natural imbalance observed in practical UAV-based blade inspection.

For experiments, CL-WTBD is split into training, validation, and test subsets with an 8:1:1 ratio, resulting in 1,754, 219, and 220 images, respectively. The split is performed at the image level. Specifically, all images are first shuffled, and stratified sampling is then applied to preserve both illumination-domain balance and defect-category proportions across the three subsets. This strategy ensures that Normal-Light, Backlight, and Low-Light images are consistently represented in training, validation, and testing, while the naturally imbalanced category distribution of Crack, Damage, and Dirt is retained.

This splitting strategy is designed to satisfy two requirements. First, because the primary objective of CL-WTBD is to evaluate illumination robustness, comparable illumination-domain distributions should be maintained across all subsets. Second, the defect categories are not artificially re-balanced, because the imbalance reflects the realistic occurrence frequency of blade defects in UAV-based inspection. As shown in Table 4, Damage is the majority class, whereas Crack is the minority class. This distribution is therefore not caused by arbitrary split bias, but by the natural category distribution retained after multi-source integration and label harmonization.

We acknowledge that image-level random splitting may still have a potential

risk of near-duplicate leakage if highly similar UAV frames from the same blade, turbine, or inspection session appear across different subsets. Because complete turbine-level, blade-level, or inspection-session metadata are not consistently available for all source datasets, a strictly session-level split cannot be fully guaranteed in the current version.

To quantify this residual risk, we conducted a perceptual hashing (pHash) audit between the training and test subsets. For each test image, we searched for its nearest neighbor in the training set using the 64-bit pHash Hamming distance. We found that 9.5% of test images have a nearest-training distance smaller than 3, and 11.4% have a distance smaller than 5, with a mean nearest-training distance of 13.96 out of 64. Manual inspection of the closest pairs indicates that these highly similar pairs mainly correspond to consecutive UAV frames captured within the same inspection burst, rather than file-level duplicates.

Since all evaluated detectors are trained and tested under the same image-level split, the relative comparison between UniIA-DET and the baselines is conducted under a controlled protocol. Nevertheless, the pHash audit indicates that the current evaluation should be interpreted as an image-level benchmark rather than a strict session-level generalization test. In future releases of CL-WTBD, we will reconstruct or collect session-level metadata and perform session-aware splitting with perceptual-hash-based duplicate screening.

To further characterize the annotation complexity of CL-WTBD, we summarize the spatial and scale distributions in Fig. 3. As shown in Fig. 3(a), each illumination domain contains samples from multiple source datasets, while the source distribution is not perfectly uniform across domains. This visualization makes the source-domain composition transparent and helps readers inspect possible source-specific biases. A quantitative source-by-domain performance analysis that examines these biases is provided in Section 5.2.4. The center-location density map in Fig. 3(b) shows that defects appear across the field of view, reflecting diverse UAV viewpoints and blade poses in real inspections. In addition, the width–height distribution in Fig. 3(c) indicates substantial variations in box scales and aspect ratios, with a clear dominance of small targets

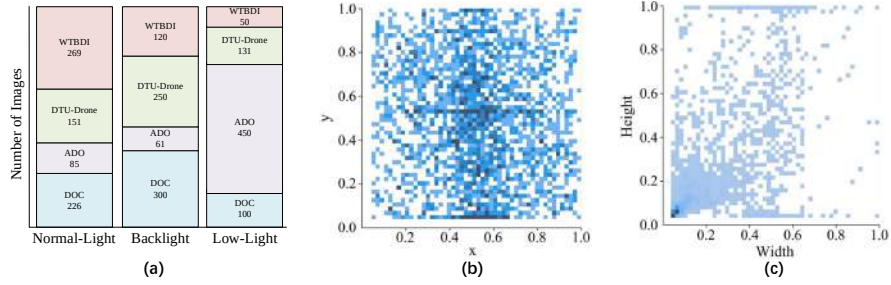


Figure 3: Dataset statistics of CL-WTBD. (a) Source-by-domain composition after illumination-domain balancing. Each illumination domain contains 731 images, while the contributions of DOC, ADO, DTU-Drone, and WTBDI differ across domains. (b) Spatial density of normalized bounding-box centers. (c) 2D density of normalized bounding-box width and height, highlighting large variations in object scales and aspect ratios.

Table 5: Sensitivity analysis after removing near-duplicate test images. pHash-clean-3 (199 images) and pHash-clean-5 (195 images) remove test images whose nearest-training pHash distance is below 3 and 5. Results are mean $\pm$ std over three seeds.

Test set	Method	P	R	Global mAP@50	Global mAP@50:95	Normal-Light	Backlight	Low-Light	Macro-domain
Original (220)	YOLOv8u	0.628 $\pm$ 0.017	0.395 $\pm$ 0.011	0.387 $\pm$ 0.020	0.242 $\pm$ 0.012	0.530 $\pm$ 0.018	0.352 $\pm$ 0.007	0.392 $\pm$ 0.012	0.424 $\pm$ 0.012
	UniIA-DET	0.675 $\pm$ 0.046	0.431 $\pm$ 0.033	0.398 $\pm$ 0.017	0.251 $\pm$ 0.015	0.558 $\pm$ 0.004	0.392 $\pm$ 0.004	0.446 $\pm$ 0.006	0.466 $\pm$ 0.002
pHash-clean-3 (199)	YOLOv8u	0.618 $\pm$ 0.017	0.389 $\pm$ 0.011	0.381 $\pm$ 0.022	0.238 $\pm$ 0.013	0.522 $\pm$ 0.021	0.345 $\pm$ 0.008	0.386 $\pm$ 0.013	0.418 $\pm$ 0.013
	UniIA-DET	0.664 $\pm$ 0.046	0.424 $\pm$ 0.033	0.392 $\pm$ 0.019	0.246 $\pm$ 0.016	0.552 $\pm$ 0.006	0.386 $\pm$ 0.006	0.441 $\pm$ 0.008	0.460 $\pm$ 0.004
pHash-clean-5 (195)	YOLOv8u	0.614 $\pm$ 0.017	0.386 $\pm$ 0.011	0.378 $\pm$ 0.023	0.236 $\pm$ 0.014	0.519 $\pm$ 0.022	0.342 $\pm$ 0.009	0.383 $\pm$ 0.014	0.415 $\pm$ 0.014
	UniIA-DET	0.659 $\pm$ 0.046	0.421 $\pm$ 0.033	0.389 $\pm$ 0.020	0.244 $\pm$ 0.017	0.549 $\pm$ 0.007	0.383 $\pm$ 0.007	0.439 $\pm$ 0.009	0.457 $\pm$ 0.005

and a non-negligible long tail of larger regions. Such heterogeneous geometries, together with low contrast under harsh illuminations, make multi-scale feature learning and bounding-box regression particularly challenging.

To test whether near-duplicate pairs inflate the results, we removed the test images whose nearest-training pHash distance is below 3 and below 5, producing two reduced test sets of 199 and 195 images, and re-evaluated both detectors. As shown in Table 5, the absolute performance decreases only slightly, while the improvement of UniIA-DET over the baseline remains essentially unchanged, with a macro-domain gain of about +0.04 in all cases. The gain remains positive under the specified pHash-based exclusion criteria. However, this analysis does not establish independence between training and test inspection sessions.

We summarize the train-test overlap along the available provenance dimensions. At the source level, all four source datasets appear in both splits by construction, since CL-WTBD is designed to assess illumination robustness on a fixed pool of sources rather than transfer to an unseen source. Turbine, blade, and session identifiers are not consistently available across the source datasets, so an exact overlap along these dimensions cannot be computed; The perceptual-hash audit measures visual similarity, and the exclusion analysis assesses sensitivity to the specified distance thresholds. Neither analysis identifies inspection sessions or quantifies all dependence between the training and test sets. A session-level generalization study, and evaluation on a fully independent external dataset, are left for future work (Section 6); a cross-source evaluation within the present source pool is reported in Section 5.2.5.

4. Methodology

For illumination-robust defect detection from UAV-captured blade images, we propose UniIA-DET, an illumination-adaptive detection framework that calibrates images in the input space before feature extraction, as illustrated in Fig. 4. The framework is built on top of an off-the-shelf object detector and focuses on improving visual robustness under complex illumination without modifying the detector topology. This design allows the proposed IAPM to be attached to different detector families in an architecture-compatible manner. Conceptually, UniIA-DET can be viewed as a lightweight calibration layer placed before the detector, where the calibration signal is not manually designed but learned from detection supervision. In this work, we adopt YOLOv8 [7] as the primary detector instantiation due to its stable reproducibility and favorable accuracy–efficiency trade-off on CL-WTBD. In this section, we first present the overview of UniIA-DET. Then, we introduce the Illumination-Adaptive Prompting Module (IAPM), followed by the detector instantiation and the joint optimization strategy.

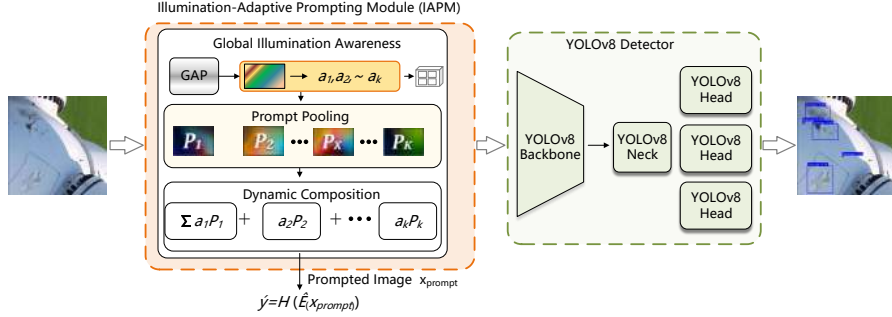


Figure 4: UniIA-DET augments an off-the-shelf object detector (e.g., YOLOv8 [7]) with an input-space IAPM. The IAPM generates a content-conditioned prompt that is injected into the raw image to form x_{prompt} , which is then processed by the detector to produce defect predictions.

4.1. Overview of UniIA-DET

As illustrated in Fig. 4, UniIA-DET augments a standard detector with an Illumination-Adaptive Prompting Module (IAPM) operated in the pixel input space. Given an input image $x \in \mathbb{R}^{H \times W \times 3}$, the IAPM generates a content-conditioned prompt and injects it additively to obtain a rectified image x_{prompt} , which is then fed into the detector:

$$\hat{y} = \mathcal{F}_{\Theta}(x + \mathcal{M}_{\phi}(x)), \quad (5)$$

where $\mathcal{M}_{\phi}$ denotes the IAPM with parameters ϕ , and $\mathcal{F}_{\Theta}$ denotes the detector with parameters Θ . In this work, we employ YOLOv8 [7] as the default instantiation. Since the detector architecture remains unchanged, UniIA-DET can leverage standard pre-trained initialization while learning illumination calibration at the earliest stage.

4.2. Illumination-Adaptive Prompting Module (IAPM)

Illumination in UAV-based inspection tasks varies across scenes and time, making a single prompt shared by all inputs inadequate. The IAPM addresses this issue by routing each input to a learnable prompt pool and composing an

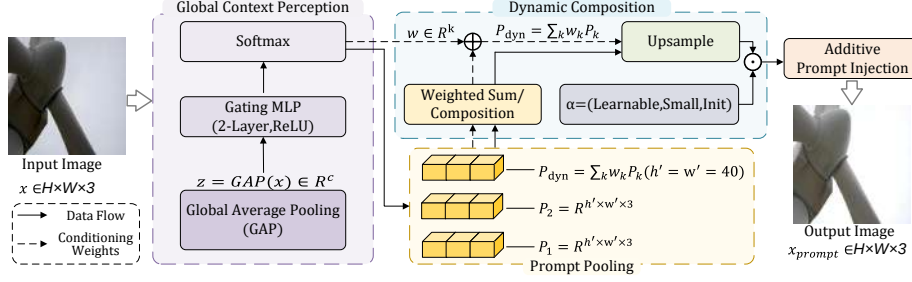


Figure 5: Architecture of the Illumination-Adaptive Prompting Module (IAPM). The module first summarizes image-level illumination statistics via Global Average Pooling (GAP) and a gating mechanism. This context generates dynamic weights $w \in \mathbb{R}^k$ to perform a weighted composition of learnable prompts $\{P_k\}$. The resulting dynamic prompt P_{dyn} is upsampled and adaptively fused with the original input x via additive injection, yielding the prompted image x_{prompt} .

input-specific correction. As illustrated in Fig. 5, the module comprises (1) a global context perception block, (2) a prompt pooling block, (3) a dynamic composition block and (4) an additive prompt injection block.

4.2.1. Global Context Perception

Since IAPM is inserted before the detector backbone, its routing descriptor is computed from the raw input image. Given an input image $x \in \mathbb{R}^{H \times W \times 3}$, we summarize its global color and brightness statistics by Global Average Pooling:

$$z = \text{GAP}(x) = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W x_{i,j}, \quad (6)$$

where $x_{i,j} \in \mathbb{R}^3$ and $z \in \mathbb{R}^3$.

The routing weights are then generated by a lightweight gating MLP:

$$w = \text{Softmax}(W_2 \sigma(W_1 z)), \quad (7)$$

where $W_1 \in \mathbb{R}^{d_g \times 3}$, $W_2 \in \mathbb{R}^{K \times d_g}$, d_g is the hidden dimension, and K is the number of prompt prototypes. We use $d_g = 16$ as the default gate width. The validation comparison conducted during this revision is reported in Section 5.4.1 and Table 15. The vector $w = \{w_1, \dots, w_K\}$ determines the contribution of each prompt prototype for the current input image.

4.2.2. Prompt Pooling and Dynamic Composition

A static visual prompt induces a fixed correction bias, which can be insufficient under the intra-class variability of outdoor illumination. We therefore maintain a small set of learnable correction prototypes and compose an input-conditioned correction from them. This formulation permits both concentrated routing to a single prototype and soft combinations of multiple prototypes. It does not, by itself, establish domain-independent representations or imply that routing is unrelated to illumination domain.

We maintain a prompt pool $\mathcal{P} = \{P_1, \dots, P_K\}$ of K learnable prototypes, where each $P_k \in \mathbb{R}^{h' \times w' \times 3}$ represents a basis correction pattern (e.g., exposure compensation or glare suppression). To control parameter growth, prototypes are stored at reduced spatial resolution (typically 40×40 pixels). The dynamic prompt is composed as:

$$P_{\text{dyn}} = \sum_{k=1}^K w_k P_k. \quad (8)$$

The nonnegative weights sum to one, so Eq. (8) defines an input-dependent convex combination of the prototypes. This formulation permits both mixed compositions and compositions dominated by one prototype. The learned routing behavior is examined in Section 5.4.7.

4.2.3. Additive Prompt Injection

Defect detection relies on fine structural cues (e.g., crack edges and subtle textures). Unrestricted pixel replacement can distort high-frequency details and destabilize downstream feature extraction. We therefore adopt a residual injection scheme, where the prompt acts as an additive correction term. Together with a learnable scaling factor, this design provides controllable calibration of illumination statistics while largely preserving geometric content.

The composed prompt is upsampled to match the input resolution and injected through a residual connection:

$$\mathbf{x}_{\text{prompt}} = \text{clip}(\mathbf{x} + \alpha \cdot \text{Upsample}(\mathbf{P}_{\text{dyn}}), 0, 1), \quad (9)$$

Algorithm 1 IAPM forward pass

Require: input image $\mathbf{x} \in \mathbb{R}^{3 \times H \times W}$; prompt pool $\{\mathbf{P}_1, \dots, \mathbf{P}_K\}$, $\mathbf{P}_k \in \mathbb{R}^{3 \times h \times w}$;
gating weights $\mathbf{W}_1 \in \mathbb{R}^{d_g \times 3}$, $\mathbf{W}_2 \in \mathbb{R}^{K \times d_g}$; scaling factor α

Ensure: corrected image $\mathbf{x}_{\text{prompt}}$

- 1: $\mathbf{z} \leftarrow \text{GAP}(\mathbf{x})$ $\triangleright$ global RGB means, $\mathbf{z} \in \mathbb{R}^3$ (Eq. 6)
 - 2: $\mathbf{a} \leftarrow \mathbf{W}_2 \sigma(\mathbf{W}_1 \mathbf{z})$ $\triangleright$ gating MLP
 - 3: $\mathbf{w} \leftarrow \text{Softmax}(\mathbf{a})$ $\triangleright$ routing weights, $\mathbf{w} \in \mathbb{R}^K$ (Eq. 7)
 - 4: $\mathbf{P}_{\text{dyn}} \leftarrow \sum_{k=1}^K w_k \mathbf{P}_k$ $\triangleright$ dynamic composition (Eq. 8)
 - 5: $\mathbf{U} \leftarrow \text{Upsample}(\mathbf{P}_{\text{dyn}})$ $\triangleright$ bilinear upsampling to $H \times W$
 - 6: $\mathbf{x}_{\text{prompt}} \leftarrow \text{clip}(\mathbf{x} + \alpha \cdot \mathbf{U}, 0, 1)$ $\triangleright$ injection + clipping (Eq. 9)
 - 7: **return** $\mathbf{x}_{\text{prompt}}$
-

where α is a learnable scalar initialized to 0.01 and $\text{Upsample}(\cdot)$ denotes bilinear interpolation. The scalar α is left unconstrained during training, while the composition weights are normalized by a softmax, so $\mathbf{P}_{\text{dyn}}$ is a convex combination of the prototypes. After injection, $\mathbf{x}_{\text{prompt}}$ is clipped to the valid image range $[0, 1]$ before the detector’s input normalization, which bounds the corrected image. This injection shifts the pixel distribution toward conditions that facilitate downstream feature extraction without introducing geometric warping.

Algorithm 1 summarizes the forward pass of the IAPM, combining the global context perception, prompt pooling, dynamic composition, and additive injection steps described in Eq. (6)–(9).

4.3. Default Detector Instantiation: The YOLOv8 Architecture

To instantiate the detector $\mathcal{F}_\Theta$, we adopt YOLOv8 [50], a state-of-the-art one-stage detector. It features a CSP-style backbone (C2f modules), an FPN/PAN feature fusion neck, and an anchor-free decoupled head. In our framework, it serves as the feature extraction backend. While effective under normal conditions, standard YOLOv8 processes raw pixel intensities directly. Under extreme backlight or low-light conditions, the signal-to-noise ratio of

the input degrades sharply, which may suppress subtle defect cues or trigger background-induced false positives. By placing the IAPM before the detector, we ensure that YOLOv8 operates on stabilized features.

4.4. Integration and Holistic Optimization

Within UniIA-DET, the IAPM is strictly inserted *before* the detector backbone. This choice targets the module where illumination degradation is most destructive, i.e., low-level primitives such as edges, micro-textures, and local contrast. Deferring correction to deeper layers risks losing high-frequency cues to early downsampling. Input-space calibration thus supplies a stabilized pixel distribution to the first feature extraction layer and improves early feature formation under backlight and low-light conditions.

The training operation is driven solely by the detection objective. We do not introduce reconstruction losses, which avoids requiring paired normal-light reference images and prevents optimization from drifting toward perceptual appearance criteria. With YOLOv8 as the default instantiation, we adopt full-parameter fine-tuning for both the detector and the IAPM. The total loss follows the standard YOLOv8 formulation:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cls}} \mathcal{L}_{\text{cls}} + \lambda_{\text{box}} \mathcal{L}_{\text{box}} + \lambda_{\text{dff}} \mathcal{L}_{\text{dff}}, \quad (10)$$

where $\mathcal{L}_{\text{cls}}$ is the classification loss, $\mathcal{L}_{\text{box}}$ uses IoU-based regression loss (e.g., CIoU) for accurate localization, and $\mathcal{L}_{\text{dff}}$ applies Distribution Focal Loss to refine box boundaries, which is beneficial when defect contours are ambiguous under low-light noise.

Given training pairs $\{(x_i, y_i)\}_{i=1}^N$, detector parameters Θ and prompting parameters ϕ are optimized jointly:

$$\Theta^*, \phi^* = \arg \min_{\Theta, \phi} \sum_{i=1}^N \mathcal{L}_{\text{total}} \left(\mathcal{F}_{\Theta} (x_i + \mathcal{M}_{\phi}(x_i)), y_i \right). \quad (11)$$

Under this objective, the prompts are shaped directly by detection gradients and are optimized toward improving classification and localization performance under illumination degradation.

4.5. Complexity Analysis

UniIA-DET is designed to be efficient for UAV-based inspection pipelines. The context gating branch is a lightweight two-layer MLP that maps the 3-dimensional global illumination descriptor to K routing weights through a small hidden layer, i.e., $3 \rightarrow d_g \rightarrow K$. The prompt pool contains K learnable RGB correction prototypes at a spatial resolution of 40×40 . We use $K = 3$ and $d_g = 16$ as the default configuration; the validation comparison conducted during this revision supports retaining these values (Section 5.4.1). At an input resolution of 640×640 , IAPM adds 0.0145 M parameters and 0.012317 GFLOPs. For YOLOv8n, the total parameter count increases from 3.011 M to 3.026 M, and the computational cost increases from 8.196 to 8.208 GFLOPs. The standalone parameter and arithmetic-cost increments are the same across the evaluated detector families when the same IAPM configuration is used. Measured runtime overhead depends on the detector architecture and implementation.

5. Experiments

Having described the UniIA-DET framework, we now evaluate its effectiveness on CL-WTBD. We conducted comprehensive experiments to assess robustness and efficiency under severe illumination shifts. Our study demonstrated that input-space prompting improves detection robustness and delivers competitive performance compared to representative methods. We further quantify the contribution of individual components through ablation studies and examine architecture compatibility by integrating UniIA-DET into diverse detection architectures.

5.1. Experimental Settings

In this subsection, we will introduce the baselines, evaluation metrics and implementation details utilized in our experiments.

5.1.1. Baselines

We benchmarked UniIA-DET against representative detectors spanning diverse architectural paradigms. All baselines were trained and evaluated on CL-WTBD according to the protocol described above.

- **One-stage CNNs:** SSD [21], YOLOv8n [50], and YOLOv11n [51]. We adopted **YOLOv8n** as the primary baseline and the default instantiation of UniIA-DET, motivated by its favorable accuracy–throughput trade-off on our benchmark.
- **Two-stage CNN:** Faster R-CNN [8] with a ResNet-50 [52] backbone, representing anchor-based two-stage detectors with strong localization capability.
- **Transformer-based:** RT-DETR [13], representing real-time detection transformers based on global attention.
- **SSM-based:** Mamba-YOLO [14], representing recent State Space Model (SSM) designs with linear-complexity sequence modeling.

5.1.2. Evaluation Metrics

We evaluated both detection quality and deployment efficiency using standard COCO-style metrics and runtime-related indicators.

We reported Precision (P) and Recall (R) to quantify the false-alarm rate and miss rate, respectively. We further reported mAP@50 (IoU = 0.5) and mAP@50:95 (mean AP over IoU thresholds from 0.5 to 0.95 in steps of 0.05). The latter emphasizes localization strictness and better reflects performance under tighter overlap criteria.

We further distinguish three levels of mAP aggregation used throughout this work. Unless otherwise stated, the mAP of an image set is the mean of the per-class AP over the three classes (Damage, Dirt, Crack). The global mAP@50 is computed over the entire test set. The per-domain mAP@50 for each illumination domain is computed on the images of that domain only, again

as the mean of the per-class AP. The macro-domain mAP@50 is the unweighted mean of the three per-domain values; it therefore averages over classes within each domain and then over domains, giving equal weight to each illumination condition. This differs from the global mAP@50, which is computed over the pooled test set and is thus affected by the number of images and instances in each domain.

$$\text{mAP}_{\text{global}} = \frac{1}{C} \sum_{c=1}^C \text{AP}_c^{\text{all}}, \quad (12)$$

$$\text{mAP}_d = \frac{1}{C} \sum_{c=1}^C \text{AP}_c^{(d)}, \quad d \in \{\text{Normal-Light, Backlight, Low-Light}\}, \quad (13)$$

$$\text{mAP}_{\text{macro}} = \frac{1}{D} \sum_{d=1}^D \text{mAP}_d, \quad (14)$$

where $C = 3$ is the number of classes, $D = 3$ is the number of illumination domains, AP_c^{all} is the AP of class c over all test images, and $\text{AP}_c^{(d)}$ is the AP of class c over the images of domain d .

Detection accuracy and illumination robustness are the primary evaluation criteria in this work. We also report the number of parameters (Params), computational cost in GFLOPs, and inference throughput in Frames Per Second (FPS) as secondary efficiency indicators. These indicators are used to quantify the relative computational overhead introduced by different modules under the same evaluation environment, rather than to claim completed onboard real-time UAV deployment.

All efficiency metrics were measured under a single unified protocol. Each model was profiled on an exclusive, idle NVIDIA GeForce RTX 3090 GPU with no concurrent jobs, using PyTorch 2.4.0 (CUDA 12.1) in FP32, with a single input tensor of shape $[1, 3, 640, 640]$. We timed the fused raw forward pass only, excluding data loading, preprocessing, post-processing, and NMS. Timing used CUDA events with `torch.cuda.synchronize()` before and after each measured region. For each model we ran 30 warm-up iterations followed by 200 timed

iterations per round over 3 rounds; for each round the FPS was computed as $1000 / \text{median latency (ms)}$, and we report the mean $\pm$ standard deviation across the 3 rounds. The number of parameters was counted on the unfused model, and GFLOPs were computed as $\text{MACs} \times 2$ using THOP, following the Ultralytics convention. For models augmented with IAPM, the reported GFLOPs are decomposed as the detector FLOPs plus the standalone IAPM FLOPs measured under the same input resolution.

5.1.3. Implementation Details

All experiments were implemented in PyTorch 2.4.0 and executed on a workstation with a single NVIDIA GeForce RTX 3090 GPU (24 GB VRAM). Unless otherwise stated, we followed a unified training protocol for all methods.

All images were resized to 640×640 pixels. We applied standard augmentations during the training process, including Mosaic, random HSV jitter, and random scaling, to reduce overfitting and improve robustness to photometric and geometric perturbations commonly observed in UAV imagery.

All models were trained for 300 epochs with a batch size of 32. To accommodate different optimization characteristics across model families, we used: (1) SGD [53] for CNN-based detectors (YOLO series [7], SSD [21], Faster R-CNN [8]) and the SSM-based Mamba-YOLO [14], with an initial learning rate of 1×10^{-2} , a momentum value of 0.937, and a weight decay of 5×10^{-4} ; and (2) AdamW [54] for the Transformer-based RT-DETR [13], with a weight decay of 1×10^{-4} for stable training.

FPS was measured under the unified efficiency protocol described in Section 5.1.2. For Zero-DCE + YOLOv8n, the timed forward pass includes the Zero-DCE enhancement front-end. FPS is therefore used only as a relative efficiency indicator under a controlled environment.

For reproducibility, we further specify the key hyperparameters of IAPM and the detector. The residual scaling factor α is initialized to 0.01. Across the three seeds, the learned α converges to 0.0087, 0.0015, and 0.0059 (0.0054 ± 0.0036), remaining small and positive; α is not otherwise constrained, and x_{prompt} is

clipped to $[0, 1]$ before detector normalization. The YOLOv8 detection loss follows the default Ultralytics setting, with $\lambda_{\text{box}} = 7.5$, $\lambda_{\text{cls}} = 0.5$, and $\lambda_{\text{diff}} = 1.5$. The prompt prototype resolution is set to 40×40 , and the number of prototypes is $K = 3$. During evaluation, the confidence threshold is set to 0.001 for PR-curve computation, and the NMS IoU threshold is set to 0.6. YOLO-based detectors are initialized from the Ultralytics COCO-pretrained weights. Unless otherwise stated, the single-run results use a fixed random seed of 0. For the repeated-run comparison in Table 6, both the fully trained YOLOv8n baseline and UniIA-DET were trained for 300 epochs under identical experimental settings using the random seeds 0, 42, and 3407.

During this revision, we conducted an additional hyperparameter comparison on the 219-image validation set. Each configuration was trained with seeds 0, 42, and 3407, and its final-epoch checkpoint was evaluated. For each seed, macro-domain mAP@50 was calculated as the arithmetic mean of the three illumination-domain scores. We report the mean and sample standard deviation across seeds. The pool-size sweep used $K \in \{1, 2, 3, 5, 7\}$ with $d_g = 16$, and the gate-width sweep used $d_g \in \{4, 8, 16, 32, 64\}$ with $K = 3$. The shared configuration $K = 3$, $d_g = 16$ uses the same three runs in both sweeps. The resulting validation values are reported in Table 15.

5.2. Experimental Results

We evaluated UniIA-DET against representative detectors on the CL-WTBD test set under the unified protocol described in Section 5.1. The baselines cover two one-stage CNN detectors (SSD [21], YOLOv11n [51]), a two-stage detector (Faster R-CNN [8]), a Transformer-based detector (RT-DETR [13]), and an SSM-based detector (Mamba-YOLO [14]).

5.2.1. Quantitative evaluation

Table 6 and 7 reports accuracy metrics (P, R, mAP@50, mAP@50:95) together with efficiency indicators (Params, GFLOPs, FPS). Averaged over three seeds, UniIA-DET reaches $P=0.675$, $R=0.431$, $mAP@50=0.398$, and

Table 6: Repeated-run stability analysis of the YOLOv8n baseline and UniIA-DET under three random seeds.

Model	Seed	P	R	mAP@50	mAP@50:95	Normal-Light	Backlight	Low-Light	Macro-domain
YOLOv8n	0	0.646	0.402	0.410	0.256	0.535	0.354	0.403	0.431
	42	0.612	0.400	0.372	0.233	0.545	0.357	0.393	0.432
	3407	0.625	0.382	0.380	0.237	0.510	0.343	0.380	0.411
	Mean $\pm$ Std	0.628 ± 0.017	0.395 ± 0.011	0.387 ± 0.020	0.242 ± 0.012	0.530 ± 0.018	0.352 ± 0.007	0.392 ± 0.012	0.424 ± 0.012
UniIA-DET	0	0.711	0.468	0.416	0.267	0.558	0.392	0.446	0.465
	42	0.623	0.409	0.395	0.242	0.562	0.388	0.452	0.467
	3407	0.690	0.415	0.382	0.242	0.555	0.395	0.441	0.464
	Mean $\pm$ Std	0.675 ± 0.046	0.431 ± 0.033	0.398 ± 0.017	0.251 ± 0.015	0.558 ± 0.004	0.392 ± 0.004	0.446 ± 0.006	0.466 ± 0.002

mAP@50:95=0.251. Compared with the YOLOv8n [7] baseline (P=0.628, R=0.395, mAP@50=0.387, mAP@50:95=0.242), UniIA-DET improves Precision by +4.7 and Recall by +3.6 percentage points; the mean paired global mAP@50 difference (+0.0103) is within the seed-to-seed variation reported in Table 6 and is therefore not the main basis of our claim. As analyzed in Section 5.2.4, the more substantial and consistent improvements appear under the backlight and low-light domains.

As shown in Table 6, we report the per-seed global and per-domain results for both the YOLOv8n baseline and UniIA-DET under three random seeds. The global mAP@50 is 0.398 ± 0.017 for UniIA-DET and 0.387 ± 0.020 for the baseline, so the global gain is within the seed-to-seed variation and is not used as our main evidence. Under Backlight and Low-Light, UniIA-DET improves over the baseline by +0.040 and +0.054 mAP@50, respectively. The paired differences are positive across all three random seeds. In the seed-0 bootstrap analysis, the Backlight and Low-Light intervals exclude zero under image-level resampling. The corresponding numerical intervals are reported in Table 11(b). Under pHash-based cluster resampling, the Backlight interval includes zero, whereas the Low-Light interval remains above zero. We therefore report these results as resampling sensitivity analyses, separately from variation across training seeds (Section 5.2.4). The global and per-domain metrics are evaluated in separate runs; the per-domain baseline results reported here are obtained from the same fully trained checkpoint used for the global evaluation. Because repeated-run

Table 7: The accuracy of YOLOv8n and UniIA-DET is averaged over three seeds; the other detectors are reported from a single run. GFLOPs are computed as MACs $\times 2$; FPS is the mean $\pm$ std over three rounds under the unified protocol (Section 5.1.2).

Method	Type	P	R	mAP@50	mAP@50:95	Params (M)	GFLOPs	FPS
Faster R-CNN	Two-stage	0.312	0.241	0.186	0.085	41.309	267.869	43.32 $\pm$ 0.74
SSD	One-stage	0.489	0.131	0.163	0.073	24.013	61.055	130.69 $\pm$ 2.05
RT-DETR	Transformer	0.513	0.261	0.205	0.112	32.812	108.001	23.20 $\pm$ 0.40
Mamba-YOLO	SSM	0.544	0.279	0.302	0.170	5.986	13.614	38.40 $\pm$ 0.50
YOLOv8n (Baseline)	One-stage	0.628	0.395	0.387	0.242	3.011	8.196	131.94 $\pm$ 1.87
UniIA-DET (Ours)	One-stage	0.675	0.431	0.398	0.251	3.026	8.208	123.16 $\pm$ 1.72

uncertainty is not exhaustively reported for the remaining baselines, this analysis is used as complementary stability evidence rather than a formal hypothesis test.

The remaining baselines exhibit obvious limitations on CL-WTBD. Both *SSD* [21] and *Faster R-CNN* [8] produced low mAP@50 scores (0.163 and 0.186), indicating insufficient robustness to harsh illuminations and low-SNR defect patterns. *RT-DETR* [13] further drops to 0.205 mAP@50 with higher computational cost (108.001 GFLOPs) and the slowest speed (23.20 $\pm$ 0.40 FPS). *Mamba-YOLO* [14] outperformed its counterparts (0.302 mAP@50) and remained efficient (38.40 $\pm$ 0.50 FPS), yet it still trailed the YOLOv8n [7] baseline and UniIA-DET in accuracy. Notably, *YOLOv11n* [51] underperformed YOLOv8n on CL-WTBD (0.276 vs. 0.387 mAP@50), suggesting that architectural updates alone do not guarantee better transfer to industrial inspection under complex lighting conditions.

5.2.2. Comparison with an enhance-then-detect baseline

We compare IAPM with two alternative input-space correction methods under the same YOLOv8n detector and detection loss. The first is Tiny Residual Correction, a fixed front-end with a budget comparable to IAPM but no prompt pool or routing: it downsamples the input to 40×40 , applies three 3×3 convolutions ($3 \rightarrow 32 \rightarrow 32 \rightarrow 3$, with SiLU and a final zero-initialized tanh), upsamples the residual to the input resolution, and injects it as $x_{\text{corr}} = \text{clip}(x +$

Table 8: Comparison with an equal-budget correction baseline and an enhance-then-detect baseline on CL-WTBD (mean $\pm$ standard deviation over three seeds). Tiny Residual Correction is a fixed convolutional correction with a budget comparable to IAPM but no prompt pool or routing; Zero-DCE is inserted as an online enhancement front-end and jointly optimized with YOLOv8n under the same detection loss. GFLOPs are computed as MACs $\times 2$, and FPS is measured under the unified protocol in Section 5.1.2. The best result in each column is shown in bold.

Method	P	R	Global mAP@50	Global mAP@50:95	Normal- Light	Backlight	Low- Light	Macro- domain	Params (M)	GFLOPs	FPS
YOLOv8n (baseline)	0.628 $\pm$ 0.017	0.395 $\pm$ 0.011	0.387 $\pm$ 0.020	0.242 $\pm$ 0.012	0.530 $\pm$ 0.018	0.352 $\pm$ 0.007	0.392 $\pm$ 0.012	0.424 $\pm$ 0.012	3.011	8.196	131.94 $\pm$ 1.87
Tiny Residual Correction	0.648 $\pm$ 0.021	0.412 $\pm$ 0.014	0.392 $\pm$ 0.012	0.246 $\pm$ 0.010	0.541 $\pm$ 0.015	0.369 $\pm$ 0.008	0.420 $\pm$ 0.013	0.443 $\pm$ 0.012	3.022	8.236	129.00 $\pm$ 1.80
Zero-DCE + YOLOv8n	0.662 $\pm$ 0.037	0.411 $\pm$ 0.017	0.395 $\pm$ 0.015	0.246 $\pm$ 0.010	0.544 $\pm$ 0.010	0.363 $\pm$ 0.010	0.434 $\pm$ 0.012	0.447 $\pm$ 0.010	3.99	73.077	58.50 $\pm$ 2.00
UniIA-DET (IAPM)	0.675 $\pm$ 0.046	0.431 $\pm$ 0.033	0.398 $\pm$ 0.017	0.251 $\pm$ 0.015	0.558 $\pm$ 0.004	0.392 $\pm$ 0.004	0.446 $\pm$ 0.006	0.466 $\pm$ 0.002	3.026	8.208	123.16 $\pm$ 1.72

$\beta \cdot r, 0, 1)$, where $\beta = 0.05 \cdot \tanh(\beta_{\text{raw}})$ is a learnable scalar. The second is a Zero-DCE-based enhance-then-detect control, in which the Zero-DCE architecture is inserted as an online enhancement front-end before YOLOv8n and the whole cascade is optimized end-to-end on CL-WTBD with the same detection loss. Both baselines are trained under the same three seeds.

As shown in Table 8, two comparisons can be drawn. At an equal budget (about 8 GFLOPs), the fixed Tiny Residual Correction improves over the plain baseline but reaches only an intermediate level: its macro-domain mAP@50 (0.443) lies between the baseline (0.424) and UniIA-DET (0.466), and under harsh illumination its backlight and low-light mAP@50 (0.369 and 0.420) remain below UniIA-DET (0.392 and 0.446). Table 8 shows higher macro-domain mAP@50 for each learned correction method than for the plain detector. However, Tiny Residual Correction and IAPM belong to different correction families: the former predicts image-dependent residuals through convolutional operations, whereas IAPM composes correction patterns from a learnable prompt pool through routing. Therefore, this comparison alone does not isolate the effect of routing. Compared with the much heavier Zero-DCE front-end, which requires 73.077 GFLOPs (about nine times the cost of IAPM), UniIA-DET reaches a higher backlight mAP@50 (0.392 vs. 0.363) and a comparable or slightly higher low-

Table 9: Per-class detection performance on CL-WTBD (mean $\pm$ standard deviation over three seeds). UniIA-DET is compared with the YOLOv8n baseline for each defect class. “Test Inst.” is the number of annotated instances of that class in the test set.

Class	Method	Test Inst.	P	R	mAP@50	mAP@50:95
Damage	YOLOv8n	286	0.705 ± 0.018	0.516 ± 0.025	0.477 ± 0.018	0.307 ± 0.014
	UniIA-DET		0.726 ± 0.022	0.557 ± 0.030	0.492 ± 0.016	0.310 ± 0.013
Dirt	YOLOv8n	88	0.650 ± 0.022	0.452 ± 0.020	0.475 ± 0.013	0.285 ± 0.007
	UniIA-DET		0.690 ± 0.033	0.515 ± 0.031	0.473 ± 0.017	0.291 ± 0.006
Crack	YOLOv8n	44	0.526 ± 0.055	0.196 ± 0.040	0.236 ± 0.035	0.149 ± 0.025
	UniIA-DET		0.615 ± 0.065	0.229 ± 0.045	0.265 ± 0.040	0.172 ± 0.030

light mAP@50 (0.446 vs. 0.434) at roughly one-ninth of the computational cost. Accordingly, this comparison supports the usefulness of learned input-space correction but does not by itself isolate the contribution of routing; the routing-specific evidence is provided by the controlled ablation in Section 5.4.8.

We emphasize that this comparison is not evidence against image enhancement. The Zero-DCE front-end itself improves over the plain baseline on every metric, raising the global mAP@50 from 0.387 to 0.395 and the macro-domain mAP@50 from 0.424 to 0.447, which is consistent with the general expectation that learned enhancement can benefit detection. Our point is narrower: for this benchmark, a detection-oriented calibration optimized directly for classification and localization attains a more favorable accuracy–efficiency trade-off than a generic enhancement front-end, reaching higher harsh-illumination accuracy at roughly one-ninth of the computational cost.

5.2.3. Class-wise performance analysis

Table 9 analyzes whether category distribution affects detection performance. The results show that class-wise performance is related to both the number of category instances and the visual characteristics of defects. Damage has the largest number of test instances and achieves the highest performance under both YOLOv8n and UniIA-DET. In contrast, Crack contains only 44 test instances

and obtains the lowest recall and AP. This is reasonable because crack defects are usually thin, low-contrast, and sensitive to illumination degradation. Therefore, missed detections of only a few crack instances can noticeably affect its class-wise recall and AP.

Compared with the YOLOv8n baseline, UniIA-DET improves the overall Precision from 0.628 to 0.675 and the Recall from 0.395 to 0.431 (mean over three seeds), while the global mAP@50 changes from 0.387 to 0.398 and the mAP@50:95 from 0.242 to 0.251, which is within the seed-to-seed variation discussed in Section 5.2.1. The benefit is more pronounced for the minority Crack class: UniIA-DET improves its Precision from 0.526 to 0.615, Recall from 0.196 to 0.229, mAP@50 from 0.236 to 0.265, and mAP@50:95 from 0.149 to 0.172. These results suggest that illumination-adaptive input correction is beneficial for enhancing weak defect cues, with the largest gains on the hardest class. However, Crack remains substantially more difficult than Damage and Dirt, indicating that minority and subtle defect detection remains challenging under complex illumination.

Over three seeds, the Dirt mAP@50 is 0.475 ± 0.013 for the baseline and 0.473 ± 0.017 for UniIA-DET; the difference is within the run-to-run variation, so the slight decrease seen at a single seed does not correspond to a systematic effect. On the same seeds, UniIA-DET improves the Dirt Precision and Recall (0.650 to 0.690 and 0.452 to 0.515) and the Dirt mAP@50:95 (0.285 to 0.291). For Damage and Crack, the mAP@50 improves from 0.477 to 0.492 and from 0.236 to 0.265, respectively. These class-wise comparisons are descriptive. With three training seeds, they do not establish the absence of class-specific degradation.

5.2.4. Domain-wise AP analysis

The CL-WTBD test set contains 220 images and 418 defect instances. As shown in Table 10, the Crack category contains only 13–17 instances in each illumination domain. Therefore, the per-domain Crack AP may have higher variance because of the limited number of instances.

As shown in Table 6, the improvement is mainly driven by the harsh illumi-

Table 10: Per-domain and per-class instance distribution in the CL-WTBD test set.

Domain	Images	Damage	Dirt	Crack	Total
Normal-Light	74	101	31	17	149
Backlight	73	94	28	14	136
Low-Light	73	91	29	13	133
Total	220	286	88	44	418

nation domains. Averaged over three seeds, UniIA-DET improves the Backlight mAP@50 from 0.352 to 0.392 and the Low-Light mAP@50 from 0.392 to 0.446 over the YOLOv8n baseline, while the gain under Normal-Light is smaller (from 0.530 to 0.558). These improvements are consistently observed across all three seeds. However, because only three random seeds are available, the statistical evidence should be interpreted as exploratory evidence of consistency rather than definitive hypothesis testing. For the three domain-wise paired t tests reported in Table 11(a), we apply the Holm correction jointly across the three tests, and we calculate the p values from the paired differences and apply the correction before rounding. The Global mean difference and its individual t interval are reported separately as an additional estimation summary and are not part of this family. No Global p value is reported. The confidence intervals are individual intervals and are not adjusted for multiple comparisons.

To further characterize uncertainty caused by the limited test size, we performed paired bootstrap analyses with 1,000 resampling iterations. The seed-level paired estimates and the fixed-model bootstrap intervals are reported in Table 11(a) and Table 11(b), respectively. Table 11(a) reports both the original unadjusted Raw p values and the Holm-adjusted p values; the Holm correction was applied across the three domain-wise seed-level comparisons.

The image-level analysis sampled individual test images with replacement within each illumination domain. The Global analysis sampled directly from all 220 test images without domain stratification. For each bootstrap iteration, we

Table 11: Paired seed-level and fixed-model bootstrap analyses of UniIA-DET relative to YOLOv8n on CL-WTBD. In panel (a), "Raw p" is the unadjusted p value and "Holm-adjusted p" applies the Holm correction jointly across the three domain-wise tests. The Global row reports an estimation summary only; no p value is reported for it.

(a) Three-seed paired statistics

Metric	Mean paired Δ	Sample SD	Individual 95% t interval	Raw p	Holm-adjusted p
Normal-Light	+0.0283	0.0147	$[-0.0083, +0.0650]$	0.080	0.080
Backlight	+0.0403	0.0107	$[+0.0138, +0.0669]$	0.023	0.045
Low-Light	+0.0543	0.0099	$[+0.0298, +0.0788]$	0.011	0.032
Global	+0.0103	0.0112	$[-0.0174, +0.0380]$	—	—

(b) Bootstrap results for the fixed seed-0 model

Metric	Seed-0 Δ	Images	Clusters	Image-level 95% CI	Cluster-level 95% CI
Normal-Light	+0.023	74	59	$[-0.010, +0.056]$	$[-0.022, +0.068]$
Backlight	+0.038	73	59	$[+0.013, +0.063]$	$[-0.004, +0.080]$
Low-Light	+0.043	73	58	$[+0.012, +0.074]$	$[+0.003, +0.083]$
Global	+0.006	220	170	$[-0.014, +0.026]$	$[-0.024, +0.036]$

first drew one index list from the specified resampling frame and applied this same list to both detectors. Repeated image indices, ground-truth annotations, and the corresponding predictions from both detectors were retained together. We recomputed mAP@50 separately for YOLOv8n and UniIA-DET on this common resampled set and then calculated the paired difference as $\text{mAP@50}(\text{UniIA-DET}) - \text{AP@50}(\text{YOLOv8n})$.

For the cluster-level analysis, connected components were formed from the pHash graph using a Hamming-distance threshold below 5. The cluster bootstrap retained all images within each selected cluster. Both analyses used two-sided 95% percentile intervals and the NumPy PCG64 generator with resampling seed 917. These intervals describe test-resampling uncertainty conditional on the fixed seed-0 models and do not quantify training-seed variation. The macro-domain average is obtained by averaging the per-domain AP values and therefore differs from the global metrics computed over the entire test set.

Table 12: Source-by-domain mAP@50 with per-cell image and ground-truth instance counts. Performance is reported as mean $\pm$ standard deviation over three random seeds.

Domain	Source	Test images	Damage	Dirt	Crack	YOLOv8n	UniIA-DET	Δ
Normal-Light	DOC	23	32	9	6	0.510 ± 0.023	0.549 ± 0.006	+0.039
	ADO	9	13	4	2	0.538 ± 0.025	0.563 ± 0.003	+0.025
	DTU-Drone	15	21	6	3	0.493 ± 0.022	0.538 ± 0.006	+0.045
	WTBDI	27	35	12	6	0.565 ± 0.018	0.576 ± 0.003	+0.011
Backlight	DOC	30	39	11	6	0.340 ± 0.009	0.387 ± 0.005	+0.047
	ADO	6	8	3	1	0.357 ± 0.016	0.398 ± 0.014	+0.041
	DTU-Drone	25	32	9	5	0.342 ± 0.010	0.386 ± 0.005	+0.044
	WTBDI	12	15	5	2	0.398 ± 0.007	0.416 ± 0.006	+0.018
Low-Light	DOC	10	13	4	2	0.357 ± 0.013	0.425 ± 0.006	+0.068
	ADO	45	56	18	8	0.404 ± 0.014	0.455 ± 0.007	+0.051
	DTU-Drone	13	17	5	2	0.352 ± 0.012	0.421 ± 0.009	+0.069
	WTBDI	5	5	2	1	0.435 ± 0.023	0.479 ± 0.009	+0.044

The paired differences are positive across all three random seeds for every illumination domain. The improvements are larger in the Backlight and Low-Light domains, supporting that UniIA-DET mainly benefits defect detection under harsh illumination conditions.

For a source-by-domain cell containing no ground-truth instance of a given class, the AP of that class is treated as undefined and excluded from the cell-level mean rather than being assigned zero. In the present evaluation, all 12 cells contain at least one instance of each defect class; therefore, every cell-level mAP@50 is calculated over Damage, Dirt, and Crack. Because several cells contain only a small number of instances, these results are used as a descriptive robustness check rather than as precise standalone performance estimates.

To check whether the domain-wise gains are confounded by source composition, we report a source-by-domain breakdown for both detectors in Table 12. As shown in Table 12, UniIA-DET improves over the baseline in all twelve source-by-domain cells, and, with the source held fixed, the improvement tends to increase as the illumination becomes harsher. The source-by-domain results show positive mean gains across the evaluated sources and illumination domains. However, this descriptive comparison does not isolate the effect of illumination

from other differences among the image subsets. Some cells contain few test images, so the absolute value of an individual cell should be interpreted with caution; the conclusion is based on the consistent direction of the gains across all cells and the within-source trend across domains.

5.2.5. Cross-Source Generalization (*Leave-One-Source-Out*)

We evaluated cross-source performance using leave-one-source-out experiments over DOC, ADO, DTU-Drone, and WTBDI. In each fold, both models were trained on three sources and evaluated on the remaining source using seeds 0, 42, and 3407. For each held-out source, we report overall mAP@50 as the mean and sample standard deviation across seeds. Paired differences were calculated within the same source and seed before summarization. The three-seed held-out-source results are summarized in Table 13(b), whereas the original single-run illumination-domain results are retained in Table 13(a) as a descriptive comparison.

Mean paired differences are positive for all four sources, ranging from 0.0207 to 0.0333. The difference is positive for all three seeds on DOC, ADO, and DTU-Drone, and for two of the three seeds on WTBDI. On WTBDI, the seed-42 difference is -0.0040 , showing that a positive mean gain does not imply improvement in every run.

For each seed, we also calculated the unweighted average over the four held-out-source scores. Across seeds, this average is 0.3364 ± 0.0087 for YOLOv8n and 0.3637 ± 0.0035 for UniA-DET, with a paired difference of $+0.0273 \pm 0.0087$. This source average differs from the illumination-domain macro-average. The original single-run illumination-domain results are retained as descriptive results and do not provide repeated-run uncertainty for individual illumination domains. All four sources belong to the CL-WTBD source pool; performance on a fully independent external dataset remains untested.

The four LOSO folds differ substantially in size and illumination composition. UniA-DET nevertheless outperforms YOLOv8n for every held-out source. Because all four sources belong to the dataset pool used in this study, LOSO is

Table 13: Panel (a) is a descriptive single-run illumination-domain analysis. Panel (b) reports overall mAP@50 as mean $\pm$ sample SD over seeds 0, 42, and 3407. Paired differences use the same source and seed.

<i>(a) Single-run illumination-domain results</i>								
Domain		YOLOv8n	UniIA-DET	Δ				
Normal-Light		0.465	0.516	+0.051				
Backlight		0.302	0.355	+0.053				
Low-Light		0.338	0.404	+0.066				
Macro-Avg.		0.368	0.425	+0.057				

<i>(b) Three-seed held-out-source results</i>								
Source	Images	Instances	Normal	Backlight	Low-Light	YOLOv8n	UniIA-DET	Paired Δ
DOC	626	1196	226	300	100	0.3573 ± 0.0085	0.3830 ± 0.0110	$+0.0257 \pm 0.0025$
ADO	596	1138	85	61	450	0.3337 ± 0.0125	0.3630 ± 0.0082	$+0.0293 \pm 0.0047$
DTU-Drone	532	1015	151	250	131	0.3107 ± 0.0146	0.3440 ± 0.0106	$+0.0333 \pm 0.0067$
WTBDI	439	837	269	120	50	0.3440 ± 0.0161	0.3647 ± 0.0086	$+0.0207 \pm 0.0214$
Unweighted average	–	–	–	–	–	0.3364 ± 0.0087	0.3637 ± 0.0035	$+0.0273 \pm 0.0087$

interpreted as a within-dataset cross-source generalization test rather than as fully independent external validation.

Panel (a) retains the original single-run illumination-domain comparison as a descriptive result. Panel (b) reports the repeated source-level analysis. The mean paired gain is positive for all four held-out sources, but the gain is not positive in every seed and source combination. These results provide evidence of cross-source robustness within the CL-WTBD source pool, but they do not establish performance on a fully independent dataset.

5.2.6. Efficiency discussion

Compared to the YOLOv8n [7] baseline, UniIA-DET introduces 0.0145M additional parameters and 0.012317 GFLOPs computational overhead, increasing the model complexity from 3.011M to 3.026M parameters and from 8.196 to 8.208 GFLOPs. Since IAPM operates before detector feature extraction, the parameter and arithmetic-cost increments are independent of the downstream detector

architecture. Under the unified timing protocol described in Section 5.1.2, the throughput decreases from 131.94 ± 1.87 to 123.16 ± 1.72 FPS. The measured runtime overhead depends on the detector architecture and implementation and cannot be inferred from the GFLOPs increment alone. Since no stage-wise timing was performed, we do not attribute the observed FPS differences to a specific IAPM operation.

5.2.7. Qualitative comparison

Fig. 6 visualizes representative test cases covering severe illumination interference. In the hub-shadow case (column (c)), several baselines missed the low-contrast defect or responded to irrelevant structures, whereas UniIA-DET maintained a stable focus on the defect region. In the backlight silhouette case (column (d)), baseline detectors frequently failed to delineate contamination boundaries, while UniIA-DET yielded tighter boxes. Under dusk/low-light conditions (column (f)), UniIA-DET better preserved defect evidence and reduced false negatives compared with YOLOv8n and Mamba-YOLO. Overall, these results demonstrate that input-space pre-correction can be used as an effective complement to standard detectors for UAV-based wind turbine inspection under complex illumination conditions.

5.3. Architecture Compatibility Analysis

UniIA-DET is designed as an input-space pre-correction module that can be coupled with heterogeneous detectors without modifying their internal topology. To examine this architecture-compatibility property, we inserted the proposed IAPM into representative detectors, covering three CNN-based one-stage and two-stage models (SSD [21], Faster R-CNN [8], YOLOv11n [51]), a Transformer-based detector (RT-DETR [13]), and an SSM-based detector (Mamba-YOLO [14]). For each detector, we report a paired comparison between the original baseline and its IAPM-augmented counterpart under the same evaluation protocol on our CL-WTBD.

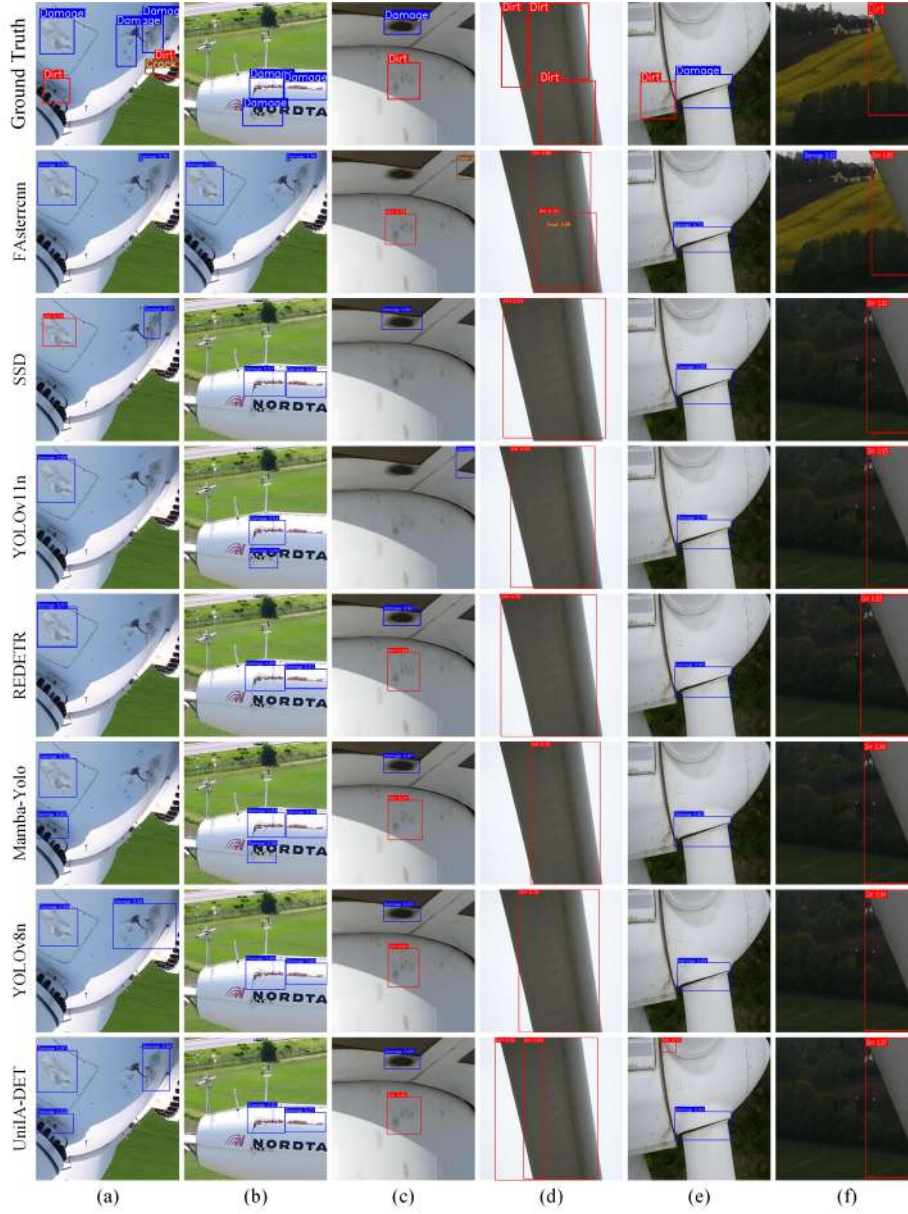


Figure 6: Qualitative comparison on CL-WTBD. The eight rows show a set of ground-truth data and seven sets of predictions of different detectors. Columns (c), (d), and (f) illustrate hub-shadow, strong backlight silhouette, and dusk/low-light cases, respectively. The blue and red boxes indicate *Damage* and *Dirt/Crack* defects, respectively. Compared to the baseline methods, UniIA-DET yields more consistent detections in the shadowed and low-illumination scenes, with fewer missed targets.

Table 14: Architecture compatibility analysis of IAPM across diverse detectors on CL-WTBD. The parameter and GFLOPs overheads denote the standalone IAPM cost measured separately from the detector backbone. Detection results for SSD, Faster R-CNN, YOLO11n, RT-DETR, and Mamba-YOLO use one training run per configuration and do not quantify training-run variability. YOLOv8n and UniIA-DET results are averaged over three seeds.

Method	P	R	mAP@50	mAP@50:95	Params (M)	GFLOPs	FPS
SSD	0.489	0.131	0.163	0.073	24.013	61.055	130.69 $\pm$ 2.05
SSD + IAPM	0.620	0.146	0.178	0.070	24.028	61.067	113.15 $\pm$ 1.89
Faster R-CNN	0.312	0.241	0.186	0.085	41.309	267.869	43.32 $\pm$ 0.74
Faster R-CNN + IAPM	0.359	0.268	0.203	0.096	41.324	267.881	42.13 $\pm$ 0.71
Mamba-YOLO	0.544	0.279	0.302	0.170	5.986	13.614	38.40 $\pm$ 0.50
Mamba-YOLO + IAPM	0.601	0.279	0.318	0.181	6.001	13.626	36.90 $\pm$ 0.60
RT-DETR	0.513	0.261	0.205	0.112	32.812	108.001	23.20 $\pm$ 0.40
RT-DETR + IAPM	0.591	0.267	0.319	0.170	32.827	108.013	22.30 $\pm$ 0.50
YOLO11n	0.553	0.268	0.276	0.145	2.590	6.443	99.88 $\pm$ 1.45
YOLO11n + IAPM	0.635	0.413	0.412	0.220	2.605	6.455	97.61 $\pm$ 1.38
YOLOv8n	0.628	0.395	0.387	0.242	3.011	8.196	131.94 $\pm$ 1.87
UniIA-DET (Ours)	0.675	0.431	0.398	0.251	3.026	8.208	123.16 $\pm$ 1.72

5.3.1. Quantitative results

Table 14 reports one training run per configuration for SSD, Faster R-CNN, YOLO11n, RT-DETR, and Mamba-YOLO. The IAPM-augmented variants have higher mAP@50 point estimates under the reported protocol. These comparisons show that IAPM can be integrated with the evaluated detector families, but they do not quantify training-run variability or establish statistically reliable gains for each architecture. The three-seed YOLOv8n/UniIA-DET comparison is reported separately in Table 6.

5.3.2. Efficiency impact

For the same IAPM configuration and an input resolution of 640×640 , IAPM adds 0.0145 M parameters and 0.012317 GFLOPs to each evaluated detector. These absolute increments in parameter count and arithmetic cost are identical across detector families. Based on the reported mean throughput, the measured

FPS reductions range from approximately 2.27% for YOLO11n to 13.42% for SSD. The runtime overhead depends on the detector architecture and implementation. Since no stage-wise timing analysis was performed, we do not attribute the observed differences to any specific IAPM operation. The FPS values represent forward-pass throughput under the hardware and timing protocol described in Section 5.1.2.

5.3.3. Qualitative evidence

Fig. 7 provides a visual, pairwise comparison between each baseline detector and its IAPM-augmented version. Across multiple challenging cases, the IAPM indicate consistent improvements in three recurring failure modes: (1) missed detections under low contrast, where predictions of the baseline were absent or fragmented (e.g., hub-region defects); (2) backlight-induced silhouette scenarios, where baselines often lost the blade boundary and produced over-extended boxes; and (3) clutter-induced false positives, where dense background patterns triggered redundant detections. In contrast, the IAPM-augmented models yielded more complete detections with cleaner box sets, supporting the quantitative improvements in both Precision and mAP.

5.4. Ablation Studies

We further conducted a series of ablation studies to isolate the contribution of each design choice. All variants adopted YOLOv8n[7] as the backbone and followed the same data split and training protocol. Those studies examined (1) the prompting formulation, (2) the training strategy, (3) the comparison with input-space normalization, (4) the accuracy–efficiency trade-off, and (5) changes in feature evidence via activation visualization.

5.4.1. Effect of the Prompting Formulation

We first compared static and *dynamic* prompting under the tuning strategies listed in Table 16. Three configurations were compared, including (1) **Baseline**: the unmodified YOLOv8n detector; (2) **Static Prompt ($K=1$)**: a single learnable tensor added to every input image, independent of the content; and

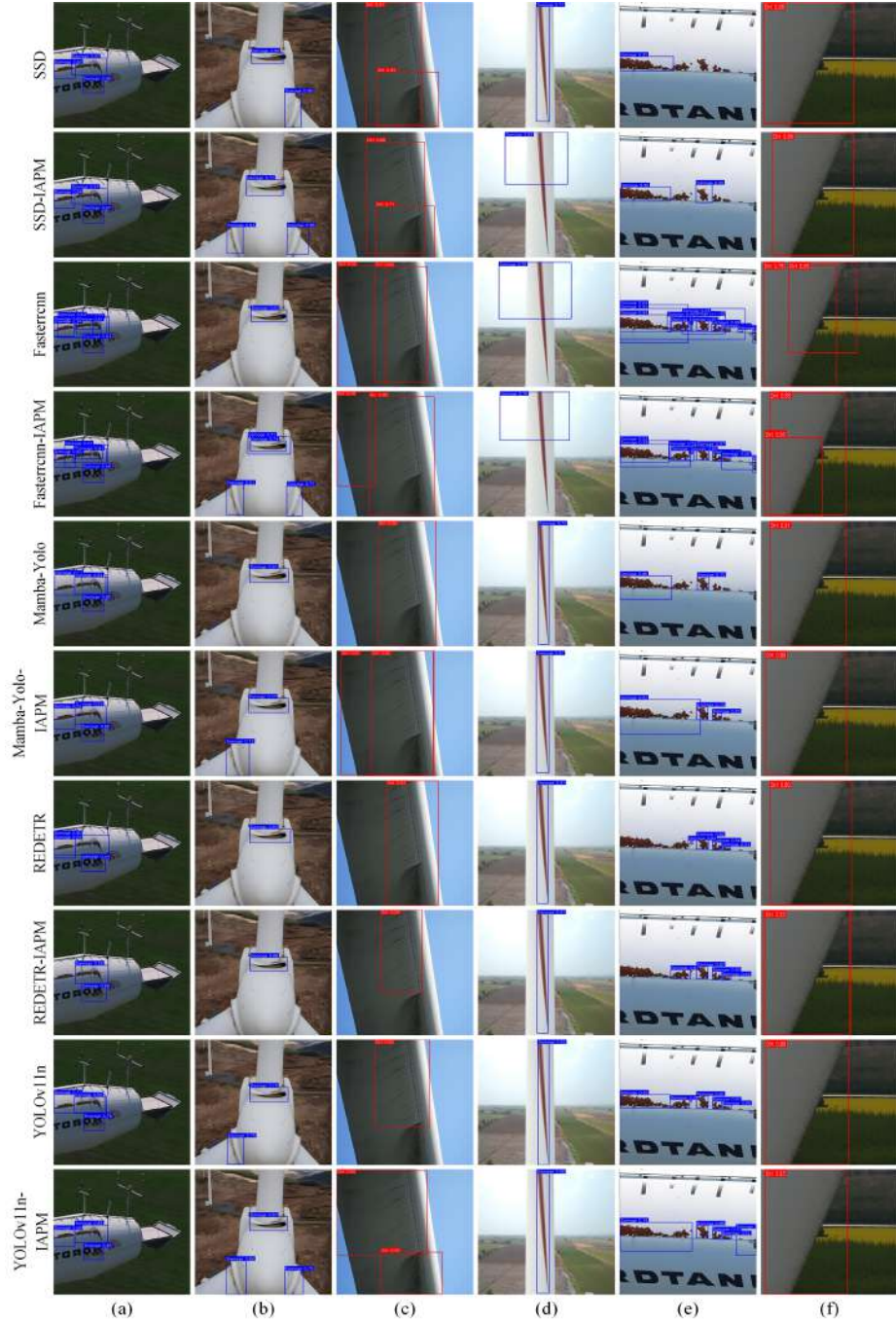


Figure 7: Qualitative examples for architecture compatibility. Each pair shows predictions of a baseline detector and its IAPM-augmented version. In the selected examples, IAPM-augmented models tend to reduce missed detections under low contrast and suppress background-induced false positives.

(3) **Dynamic Prompt ($K=3$):** the proposed IAPM, which routed among K prompt prototypes conditioned on the global context.

As reported in Table 16, the static prompt improved the Precision ($0.645 \rightarrow 0.659$) and Recall ($0.402 \rightarrow 0.425$) metrics, implying that a global additive bias can partially compensate for systematic photometric shifts. However, mAP@50 remained similar (0.410 vs. 0.409), indicating that a single fixed correction is insufficient for the heterogeneous degradations in CL-WTBD. In contrast, the dynamic prompt with full fine-tuning yielded the strongest overall results among the four configurations in Table 16. The effect of input-dependent routing is examined separately in Section 5.4.8.

Table 17 reports the effect of the prompt-pool size K . The macro-domain mAP@50 increases from $K = 1$ to $K = 3$, is similar at $K = 5$, and decreases at $K = 7$. The routing analysis in Section 5.4.7 shows three dominant prototypes at $K = 5$, with the two additional prototypes receiving low mean routing weights. The similar detection performance of $K = 3$ and $K = 5$ is compatible with this pattern and does not rule out specialization to the three illumination domains.

The existing pool-size and gate-width results in Tables 17 and 18 are test-set sensitivity comparisons. During this revision, we separately reassessed the default configuration on the validation set using the protocol described in Section 5.1.3. With $d_g = 16$, $K = 3$ achieves the highest mean validation macro-domain mAP@50, followed by $K = 5$. The difference between their mean values is 0.0007 , which is not interpreted as evidence of statistical superiority. With $K = 3$, $d_g = 16$ achieves the highest mean among the evaluated gate widths. The validation results are summarized in Table 15. These results support retaining $K = 3$, $d_g = 16$ as the default configuration; however, they do not establish that this setting is optimal over all possible combinations of the two hyperparameters.

5.4.2. Comparison with Input-Space Normalization

We compare IAPM with three near-input normalization baselines built on YOLOv8n, all placed before the backbone and supervised only by the detection loss. Per-image standardization is a fixed preprocessing step without learnable

Table 15: Validation-set sensitivity to the prompt-pool size K and gate width d_g . Results are reported as mean $\pm$ sample standard deviation over three training seeds. The shared configuration $K = 3$, $d_g = 16$ uses the same three runs in both sweeps.

(a) Prompt-pool size sweep ($d_g = 16$)					
K	1	2	3	5	7
Validation macro-domain mAP@50	0.4437 $\pm$ 0.0038	0.4600 $\pm$ 0.0026	0.4730 $\pm$ 0.0033	0.4723 $\pm$ 0.0060	0.4633 $\pm$ 0.0028
(b) Gate-width sweep ($K = 3$)					
d_g	4	8	16	32	64
Validation macro-domain mAP@50	0.4170 $\pm$ 0.0002	0.4410 $\pm$ 0.0038	0.4730 $\pm$ 0.0033	0.4593 $\pm$ 0.0029	0.4307 $\pm$ 0.0031

Table 16: Ablation on the prompt type (static vs. dynamic) and the tuning strategy (frozen vs. full), evaluated on CL-WTBD with YOLOv8n. Results are reported from a single run (seed=0); the multi-seed results for the default configuration (IAPM, Ours) are reported in Tables 5 and 6. Params and FPS are reported for the $K=3$ configuration; the static $K=1$ variant has a marginally smaller cost that does not affect the comparison. The best result in each column is shown in bold.

Method	Prompt Type	Tuning	P	R	mAP@50	mAP@50:95	Params (M)	GFLOPs	FPS
YOLOv8n	N/A	Full	0.645	0.402	0.410	0.256	3.011	8.196	131.94 $\pm$ 1.87
+ IAPM (Static)	Fixed ($K=1$)	Full	0.659	0.425	0.409	0.255	3.016	8.208	123.16 $\pm$ 1.72
+ IAPM (Dynamic)	Context ($K=3$)	Frozen	0.663	0.428	0.414	0.257	3.026	8.208	123.16 $\pm$ 1.72
+ IAPM (Ours)	Context ($K=3$)	Full	0.711	0.468	0.416	0.267	3.026	8.208	123.16 $\pm$ 1.72

parameters; Input IN-affine adds a static channel-wise affine transform after instance normalization; and Adaptive input-affine predicts channel-wise affine parameters from the global RGB means. As shown in Table 19, none of the three variants improves detection, and all fall below the plain baseline on every metric, with a drop that is also present under Normal-Light. This indicates that re-scaling the raw input by its channel statistics tends to remove input information on which the pretrained detector depends. In contrast, the bounded and spatially structured additive correction in IAPM improves over the baseline across all metrics, which shows that injecting an illumination adjustment while preserving

Table 17: Effect of the prompt-pool size K in IAPM on CL-WTBD. Results are reported as mean $\pm$ standard deviation over three seeds. $K = 3$ and $K = 5$ yield similar performance in the reported experiments. The best result in each column is shown in bold.

Method	P	R	mAP@50	mAP@50:95	Normal-Light	Backlight	Low-Light	Macro-domain
YOLOv8n	0.628 ± 0.017	0.395 ± 0.011	0.387 ± 0.020	0.242 ± 0.012	0.530 ± 0.018	0.352 ± 0.007	0.392 ± 0.012	0.424 ± 0.012
$K = 1$	0.646 ± 0.025	0.410 ± 0.017	0.389 ± 0.019	0.244 ± 0.013	0.541 ± 0.012	0.365 ± 0.010	0.406 ± 0.014	0.437 ± 0.009
$K = 2$	0.660 ± 0.034	0.422 ± 0.024	0.394 ± 0.018	0.248 ± 0.014	0.552 ± 0.007	0.379 ± 0.007	0.431 ± 0.009	0.454 ± 0.005
$K = 3$	0.675 ± 0.046	0.431 ± 0.033	0.398 ± 0.017	0.251 ± 0.015	0.558 ± 0.004	0.392 ± 0.004	0.446 ± 0.006	0.466 ± 0.002
$K = 5$	0.671 ± 0.041	0.429 ± 0.030	0.398 ± 0.018	0.251 ± 0.015	0.557 ± 0.005	0.389 ± 0.006	0.449 ± 0.006	0.465 ± 0.003
$K = 7$	0.657 ± 0.047	0.421 ± 0.034	0.394 ± 0.020	0.247 ± 0.016	0.554 ± 0.009	0.382 ± 0.011	0.438 ± 0.012	0.458 ± 0.007

Table 18: Effect of the gating-MLP hidden width d_g in IAPM on CL-WTBD. Results are reported as mean $\pm$ standard deviation over three seeds. Accuracy peaks at $d_g=16$ on all metrics while parameters and GFLOPs stay almost constant; the best result in each column is shown in bold.

d_g	P	R	mAP@50	mAP@50:95	Normal-Light	Backlight	Low-Light	Macro-domain	Params (M)	GFLOPs
4	0.544 ± 0.023	0.401 ± 0.021	0.370 ± 0.034	0.240 ± 0.003	0.539 ± 0.005	0.342 ± 0.008	0.341 ± 0.005	0.407 ± 0.027	3.026	8.208
8	0.613 ± 0.052	0.409 ± 0.026	0.382 ± 0.027	0.249 ± 0.018	0.530 ± 0.006	0.335 ± 0.002	0.368 ± 0.002	0.411 ± 0.023	3.026	8.208
16	0.675 ± 0.046	0.431 ± 0.033	0.398 ± 0.017	0.251 ± 0.015	0.558 ± 0.004	0.392 ± 0.004	0.446 ± 0.006	0.466 ± 0.002	3.026	8.208
32	0.593 ± 0.040	0.402 ± 0.053	0.382 ± 0.037	0.240 ± 0.014	0.550 ± 0.005	0.360 ± 0.006	0.379 ± 0.003	0.430 ± 0.004	3.026	8.208
64	0.549 ± 0.070	0.408 ± 0.044	0.371 ± 0.052	0.239 ± 0.034	0.544 ± 0.008	0.313 ± 0.003	0.358 ± 0.006	0.405 ± 0.049	3.026	8.208

geometric and distributional cues is more effective than direct normalization for this task.

5.4.3. Analysis of the Learned Correction

We analyze the behavior of the learned residual correction from both the magnitude and functional perspectives. The scaling factor α in Eq. 9 is a single learnable scalar initialized to 0.01 and is left unconstrained during training. Across the three seeds it converges to 0.0087, 0.0015, and 0.0059 (0.0054 ± 0.0036), remaining small and positive in every run.

We distinguish the unscaled field $U = \text{Upsample}(P_{\text{dyn}})$, the scaled residual $r = \alpha U$, and the post-clipping change $\Delta x = \text{clip}(x + r, 0, 1) - x$. The detector receives the corrected image $x' = x + \Delta x$.

Figure 8 shows the input and corrected RGB images, together with $\max_c |U_c|$, $\max_c |\alpha U_c|$, and $\max_c |\Delta x_c|$, where c indexes the RGB channels. The scaled-

Table 19: Comparison with input-space normalization baselines on CL-WTBD (YOLOv8n, three-seed mean $\pm$ std). Best result in each column in bold.

Method	P	R	mAP@50	mAP@50:95	Normal-Light	Backlight	Low-Light	Macro-domain
YOLOv8n (baseline)	0.628 $\pm$ 0.017	0.395 $\pm$ 0.011	0.387 $\pm$ 0.020	0.242 $\pm$ 0.012	0.530 $\pm$ 0.018	0.352 $\pm$ 0.007	0.392 $\pm$ 0.012	0.424 $\pm$ 0.012
Per-image standardization	0.568 $\pm$ 0.035	0.301 $\pm$ 0.014	0.313 $\pm$ 0.017	0.167 $\pm$ 0.007	0.285 $\pm$ 0.057	0.242 $\pm$ 0.014	0.339 $\pm$ 0.025	0.289 $\pm$ 0.028
Input IN-affine	0.510 $\pm$ 0.043	0.350 $\pm$ 0.037	0.325 $\pm$ 0.016	0.164 $\pm$ 0.013	0.315 $\pm$ 0.045	0.264 $\pm$ 0.034	0.336 $\pm$ 0.019	0.305 $\pm$ 0.028
Adaptive input-affine	0.571 $\pm$ 0.034	0.333 $\pm$ 0.022	0.329 $\pm$ 0.021	0.179 $\pm$ 0.016	0.460 $\pm$ 0.022	0.241 $\pm$ 0.019	0.365 $\pm$ 0.024	0.356 $\pm$ 0.007
UniIA-DET (IAPM)	0.675 $\pm$ 0.046	0.431 $\pm$ 0.033	0.398 $\pm$ 0.017	0.251 $\pm$ 0.015	0.558 $\pm$ 0.004	0.392 $\pm$ 0.004	0.446 $\pm$ 0.006	0.466 $\pm$ 0.002

Table 20: Fraction of pixels clipped after injection on CL-WTBD, per illumination domain (averaged over three seeds). Only about 1% of pixels are clipped, with a slightly higher ratio under harsher illumination.

Domain	Normal-Light	Backlight	Low-Light	Global
Clipped-pixel ratio	0.0053	0.0100	0.0139	0.0097

residual and actual-change maps share a fixed intensity scale. The input and corrected RGB images use the same display mapping without gamma or contrast enhancement. The displayed examples show small but spatially structured changes. The magnitude maps do not retain the sign of the residual and therefore do not indicate whether a region is brightened or darkened.

Clipping frequency is reported separately from correction magnitude because it also depends on the input intensity and the direction of the residual. The domain-wise clipped-pixel ratios are reported in Table 20. A higher clipping frequency does not, by itself, establish a larger correction.

Table 21 compares the independently trained YOLOv8n baseline with UniIA-DET evaluated with the learned correction enabled and disabled at inference.

Disabling the correction reduces the reported mAP@50 from 0.558 to 0.551 under Normal-Light, from 0.392 to 0.317 under Backlight, and from 0.446 to 0.404 under Low-Light. The corresponding differences between the enabled and disabled conditions are +0.007, +0.075, and +0.042. Retaining the correction benefits the evaluated, jointly trained system, with larger differences under Backlight and Low-Light. The detector was optimized on corrected inputs

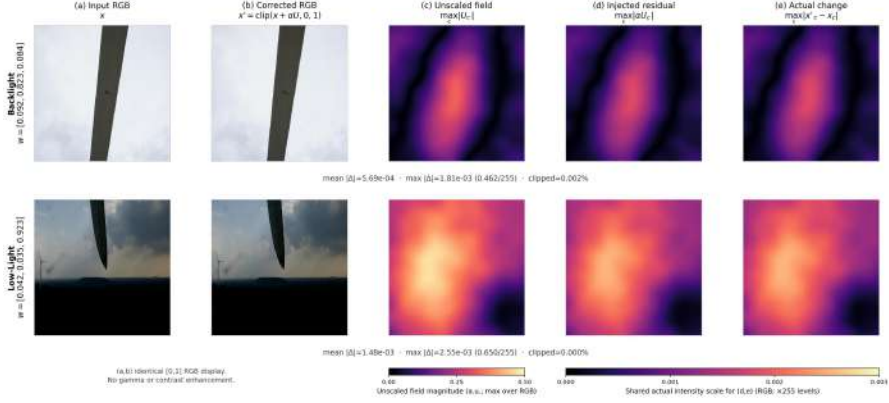


Figure 8: Magnitude of the learned correction for representative Backlight and Low-Light images. Columns show (a) the input x , (b) the corrected image $x' = \text{clip}(x + \alpha U, 0, 1)$, (c) the unscaled field magnitude $\max_c |U_c|$, (d) the scaled residual magnitude $\max_c |\alpha U_c|$, and (e) the actual post-clipping change $\max_c |x'_c - x_c|$. The unscaled field uses a separate 0–0.50 scale; columns (d) and (e) share a 0–0.003 scale in normalized RGB intensity units. The input and corrected RGB images use the same display mapping. The reported statistics describe the displayed examples.

during joint training, so disabling the correction at inference changes its input distribution. This intervention measures dependence on the correction at test time; it does not separate the correction from training-time co-adaptation or other effects of joint optimization.

The plain YOLOv8n baseline was trained independently. Setting $\alpha = 0$ only at inference does not recover that baseline. Relative to the reported baseline domain scores, the correction-disabled model differs by +0.021 in Normal-Light, −0.035 in Backlight, and +0.012 in Low-Light. In Backlight, the enabled-minus-disabled difference is 0.075, whereas the difference between the complete model and the independently trained baseline is 0.040. These comparisons measure different quantities.

5.4.4. Impact of the Optimization Strategy

To examine whether or not UniIA-DET benefits from co-adaptation between the IAPM and the detector, we compared its two variants, including (1) **Frozen**

Table 21: Effect of enabling the learned correction at inference.

Setting	Global mAP@50	Normal- Light	Backlight	Low- Light
YOLOv8n baseline	0.387 ± 0.020	0.530	0.352	0.392
UniIA-DET, correction enabled	0.398 ± 0.017	0.558	0.392	0.446
Same UniIA-DET checkpoint, $\alpha = 0$ at inference	0.391 ± 0.015	0.551	0.317	0.404
Enabled — disabled	$+0.006 \pm 0.013$	+0.007	+0.075	+0.042

Backbone: YOLOv8n was fixed and only the IAPM was optimized; and (2)

Full Fine-Tuning: YOLOv8n and IAPM were optimized jointly. The results show that the frozen setting improved mAP@50 to 0.414, exceeding both the baseline and the static prompt variant, which suggests that the proposed IAPM can serve as a drop-in front-end correction module. The full fine-tuning variant further improved performance, most notably in Precision (0.711 vs. 0.663). The comparison is consistent with a benefit from joint optimization of IAPM and the detector.

5.4.5. Efficiency and Complexity

For UAV-based inspection, throughput and memory footprint are important efficiency considerations. As shown in Table 7, the standalone IAPM introduces only 0.0145M additional parameters and 0.012317 GFLOPs computational overhead under the unified profiling protocol. This overhead is independent of the downstream detector architecture because IAPM only processes the input image before feature extraction. The measured throughput decreases from 131.94 ± 1.87 to 123.16 ± 1.72 FPS under the same evaluation protocol. The FPS change is reported as a practical runtime indicator and does not directly correspond to the FLOPs increase alone. Because we did not conduct a stage-wise timing breakdown, we do not attribute the measured throughput change to any specific IAPM operation.

5.4.6. Visualization of Activations

To provide qualitative observations of model responses under harsh illumination, we visualize the feature activations of different methods using Grad-CAM [55], as shown in Fig. 9. The Grad-CAM visualizations are used as qualitative evidence rather than mechanistic proof. In the selected examples, the YOLOv8n baseline often shows responses on high-contrast background structures under backlight and weak, scattered responses under low-light. The static prompt strengthens activation responses, but the responses may spread along blade boundaries. In comparison, the IAPM-augmented detector tends to produce more concentrated activation around defect regions and fewer responses to irrelevant background areas. These observations are consistent with the quantitative improvements, but they do not by themselves prove the internal mechanism of IAPM.

5.4.7. Routing behavior analysis

We examined the routing weights of the evaluated checkpoints with $K = 3$ and $K = 5$ on the same 220 test images. The set contained 74 Normal-Light, 73 Backlight, and 73 Low-Light images. Prototype indices are specific to each checkpoint and do not imply matched identities across configurations or independent training runs.

For each image i , we calculated the routing entropy $H_i = -\sum_{k=1}^K w_{ik} \ln w_{ik}$, its normalized value $H_i / \ln K$, and the maximum weight $\max_k w_{ik}$. We averaged these quantities over images rather than calculating entropy from the domain-average weights. For each prototype, mean routing utilization was calculated as its average soft weight across images. We also counted the images for which each prototype had the largest weight. We report raw entropy, normalized entropy, and the maximum routing weight because the normalization factor $\ln K$ varies with the prompt-pool size. Table 22 summarizes the domain-wise mean routing weights, normalized entropy, and dominant-prototype assignments for $K = 3$ and $K = 5$.

At $K = 5$, P_4 and P_5 received mean routing weights of 2.4231% and 2.2889%,

Table 22: The overall mean maximum routing weight is 0.8702 for $K = 3$ and 0.8452 for $K = 5$. Prototype identities are specific to each checkpoint.

<i>(a) Domain-wise mean routing weights</i>						
K	Domain	P_1	P_2	P_3	P_4	P_5
3	Normal-Light	0.8642	0.0715	0.0643	–	–
3	Backlight	0.0921	0.8235	0.0844	–	–
3	Low-Light	0.0418	0.0352	0.9230	–	–
5	Normal-Light	0.7797	0.2160	0.0015	0.0014	0.0014
5	Backlight	0.0830	0.8425	0.0007	0.0125	0.0613
5	Low-Light	0.0129	0.0075	0.9143	0.0591	0.0063
<i>(b) Entropy and dominant routing</i>						
K	Domain	$H / \ln K$	Dominant prototype	Dominant count	Mean $\max(w)$	
3	Normal-Light	0.3798	P_1	74/74	–	
3	Backlight	0.4424	P_2	73/73	–	
3	Low-Light	0.2475	P_3	73/73	–	
5	Normal-Light	0.3396	P_1	74/74	–	
5	Backlight	0.3532	P_2	73/73	–	
5	Low-Light	0.2276	P_3	73/73	–	

respectively, and neither was dominant for any evaluated image. The pool therefore contained three dominant prototypes and two additional prototypes with low mean routing utilization. Low routing weights alone do not establish that a prototype is functionally redundant, because its correction contribution also depends on its amplitude and spatial structure. In these checkpoints, IAPM learned image-conditioned prototype compositions with strong illumination-domain specialization. The routing statistics describe the use of the prototype pool; they do not establish a detection advantage of soft mixing over hard selection. Because routing uses globally pooled RGB statistics, this analysis concerns image-level illumination. The limitations for localized shadows and glare are discussed in Section 6.

Columns: (a) input, (b) baseline, (c) static prompt, (d) dynamic IAPM with frozen backbone, (e) **UniIA-DET**.

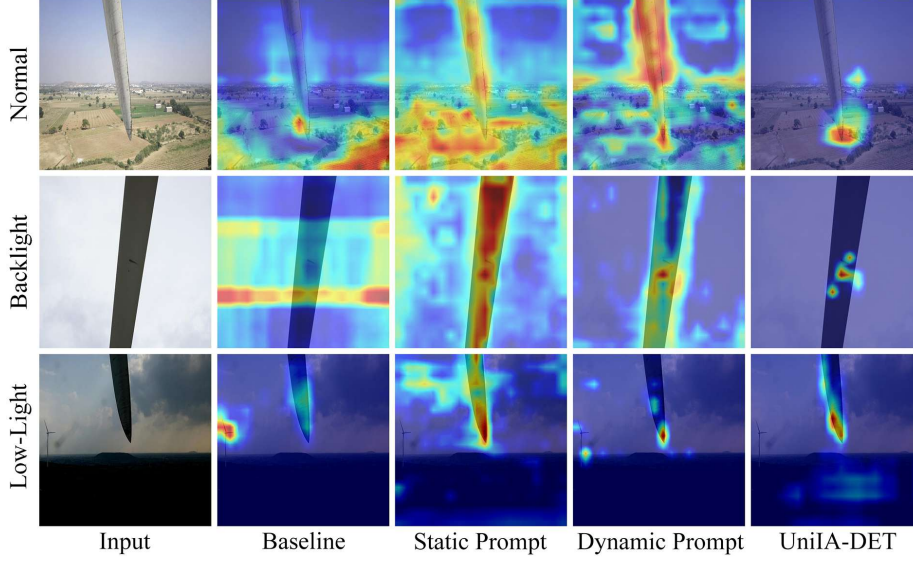


Figure 9: Grad-CAM visualizations of the YOLOv8n baseline and IAPM variants. The visualizations provide qualitative observations of activation patterns under harsh illumination. In the selected examples, the IAPM-augmented detector tends to show more concentrated responses around defect regions and fewer responses to irrelevant background areas.

5.4.8. Routing-Isolated Ablation Study

Table 23 compares Uniform Routing, Learned Fixed Mixture, and Content Routing under matched $K = 3$ settings. Uniform Routing uses fixed weights $w_k = 1/K$, Learned Fixed Mixture uses one learned weight vector shared by all images, and Content Routing predicts image-dependent weights from global RGB statistics. The baseline and single-prototype results are reported separately in Tables 6 and 16.

We report three runs per configuration using seeds 0, 42, and 3407. For each run, macro-domain mAP@50 is calculated as the arithmetic mean of the three domain scores according to Eq. (14). Table 23(a) reports the mean and sample standard deviation across seeds. Table 23(b) reports paired macro-domain differences calculated at the same seed.

Content Routing achieves a macro-domain mAP@50 of 0.4654 ± 0.0018 , compared with 0.4337 ± 0.0080 for Learned Fixed Mixture and 0.4310 ± 0.0078

Table 23: All entries are mAP@50. Macro-domain mAP@50 is the arithmetic mean of the three illumination-domain values for each seed. Results are reported as mean $\pm$ sample standard deviation across three seeds.

<i>(a) Three-seed results</i>				
Configuration	Normal-Light	Backlight	Low-Light	Macro-domain
Uniform Routing	0.5330 ± 0.0131	0.3570 ± 0.0072	0.4030 ± 0.0079	0.4310 ± 0.0078
Learned Fixed Mixture	0.5350 ± 0.0095	0.3600 ± 0.0105	0.4060 ± 0.0044	0.4337 ± 0.0080
Content Routing	0.5583 ± 0.0035	0.3917 ± 0.0035	0.4463 ± 0.0055	0.4654 ± 0.0018
<i>(b) Paired macro-domain differences</i>				
Comparison	Seed 0	Seed 42	Seed 3407	Mean $\pm$ SD
Content – Learned Fixed Mixture	+0.0343	+0.0247	+0.0363	$+0.0318 \pm 0.0062$
Content – Uniform	+0.0300	+0.0317	+0.0417	$+0.0345 \pm 0.0063$

for Uniform Routing. The paired macro-domain differences are $+0.0318 \pm 0.0062$ and $+0.0345 \pm 0.0063$, respectively. Content Routing outperforms both controls in each illumination domain for all three seeds.

These results support the benefit of input-dependent weights over fixed mixtures within the evaluated $K = 3$ IAPM configuration. They do not establish that routing accounts for most of the full system’s improvement over the unmodified detector. The comparison also does not test soft routing against hard prototype selection.

5.4.9. Failure Analysis

Although UniIA-DET improves detection robustness under harsh illumination, several failure cases remain.

On the CL-WTBD test set, the remaining false negatives mainly originate from small defects, localized illumination artifacts, and ambiguous surface patterns. In particular, small crack defects remain challenging because their spatial resolution is limited after resizing.

Moreover, the current routing mechanism relies on global illumination statis-

tics. Therefore, localized shadows or glare regions cannot always be fully characterized. Future work will explore spatially adaptive illumination modeling to address these limitations.

6. Conclusion and Future Work

In this paper, we addressed UAV-based wind turbine blade defect detection under uncontrolled illumination from both the data and the model side. We first built CL-WTBD, a complex-lighting blade defect benchmark with balanced normal-light, backlight, and low-light domains, which gives a controlled testbed for measuring illumination robustness. We then proposed UniIA-DET. It inserts the Illumination-Adaptive Prompting Module (IAPM) before the detector backbone and composes an input-specific correction from a learnable prompt pool, calibrating exposure and contrast in pixel space before feature extraction. Unlike standalone enhancement methods tuned for visual quality, IAPM is trained jointly with the detector under the detection loss, so the correction keeps defect-relevant structures instead of only making the image look clearer. On CL-WTBD, the gains appear mainly under harsh illumination, with consistent three-fold improvements under backlight and low-light (Table 6), and IAPM attaches to different detector families, including RT-DETR and Mamba-YOLO, without changing their architectures. This demonstrates compatibility with existing detector architectures under controlled offline evaluation.

Several evaluation limitations remain. CL-WTBD uses an image-level split, and incomplete turbine, blade, and inspection-session metadata prevent verified separation by inspection session. Related frames may therefore occur across subsets, and generalization to unseen inspection sessions has not been established. The pHash analyses are sensitivity checks and do not reconstruct session units. The LOSO experiments assess source shifts within the CL-WTBD source pool; performance on a fully independent external dataset remains untested. The compatibility comparisons involving SSD, Faster R-CNN, YOLO11n, RT-DETR, and Mamba-YOLO each use one training run per configuration, so their training-run

variability remains unquantified. Detection accuracy remains modest, the three macro-classes do not support fine-grained recognition, and globally conditioned correction may be less effective for localized glare or shadows. The experiments are offline and do not evaluate UAV flight-control integration or active re-capturing. We plan to extend the benchmark and evaluation to additional inspection datasets and real flight sessions for cross-dataset generalization, and to run formal multi-seed hypothesis testing across all baselines. We also intend to explore fine-grained defect recognition and severity assessment, spatially adaptive correction for local shadows and glare, lightweight deployment on UAV edge devices, and inspection workflows in which detection drives active re-capturing and viewpoint adjustment.

Conflict of Interests

The authors declare that no potential competing interests exist. There is no undisclosed relationship that may pose a competing interest, and no undisclosed funding source that may pose a competing interest.

Data availability

The metadata and derived materials for CL-WTBD, including annotation files, dataset index files, source-to-domain mappings, illumination-domain assignment rules, threshold settings, and train/validation/test split files, will be publicly released upon acceptance. DOC source images will be included as permitted by their applicable license. Original images from source datasets with redistribution restrictions will not be redistributed. For these sources, we will provide source identifiers, image indices, mappings, acquisition instructions, and scripts for reconstructing the dataset index. The UniIA-DET source code, data-processing and evaluation scripts, configuration files, training instructions, and trained weights will also be released publicly. All materials will be distributed in accordance with the applicable source-dataset licenses.

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CRediT authorship contribution statement

Xunyang Su: Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft; **Chengjia Wang:** Formal analysis, Methodology, Writing - Review & Editing; **Xinghui Dong:** Conceptualization, Funding acquisition, Investigation, Methodology, Project Administration, Resources, Supervision, Writing - Review & Editing.

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